

October 23, 2025

Matthew William Kenzie 15:39

<https://resources-old.web.cern.ch/resources-old/Manage/ListServices.aspx>
<https://pdg.lbl.gov/2025/reviews/rpp2024-rev-ckm-matrix.pdf>
<https://pdg.lbl.gov/2025/reviews/rpp2024-rev-cp-violation.pdf>
<https://pdg.lbl.gov/2025/reviews/rpp2024-rev-b-bar-mixing.pdf>
<https://pdg.lbl.gov/2025/reviews/rpp2024-rev-ckm-angles.pdf>
https://www.hep.phy.cam.ac.uk/~mkenzie/teaching/flavour/Lecture4_Kenzie.pdf
https://www.hep.phy.cam.ac.uk/~mkenzie/teaching/flavour/Lecture5_Kenzie.pdf
<https://inspirehep.net/literature/2685074>
<https://inspirehep.net/literature/1623025>
<https://hep-fcc.github.io/fcc-tutorials/cern2022/index.html#>
<https://github.com/HEP-FCC/FCCAnalyses/tree/master/examples/FCCee/tutorials>

GitHub

[FCCAnalyses/examples/FCCee/tutorials at master · HEP-FCC/FCCAnalyses](#)

[Common analysis framework for the Future Circular Collider - HEP-FCC/FCCAnalyses](#)



HEP-FCC/
FCCAnalyses



Common analysis framework for the Future Circular Collider

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October 24, 2025

Houde Li Update your status 09:23

Hi, I have few questions about the risk assessment form.

We need to fill in a divisional local safety coordinator and a location of activity.

Matthew William Kenzie 09:24

location is RDC

no idea who the divisionla local safety coordinator is

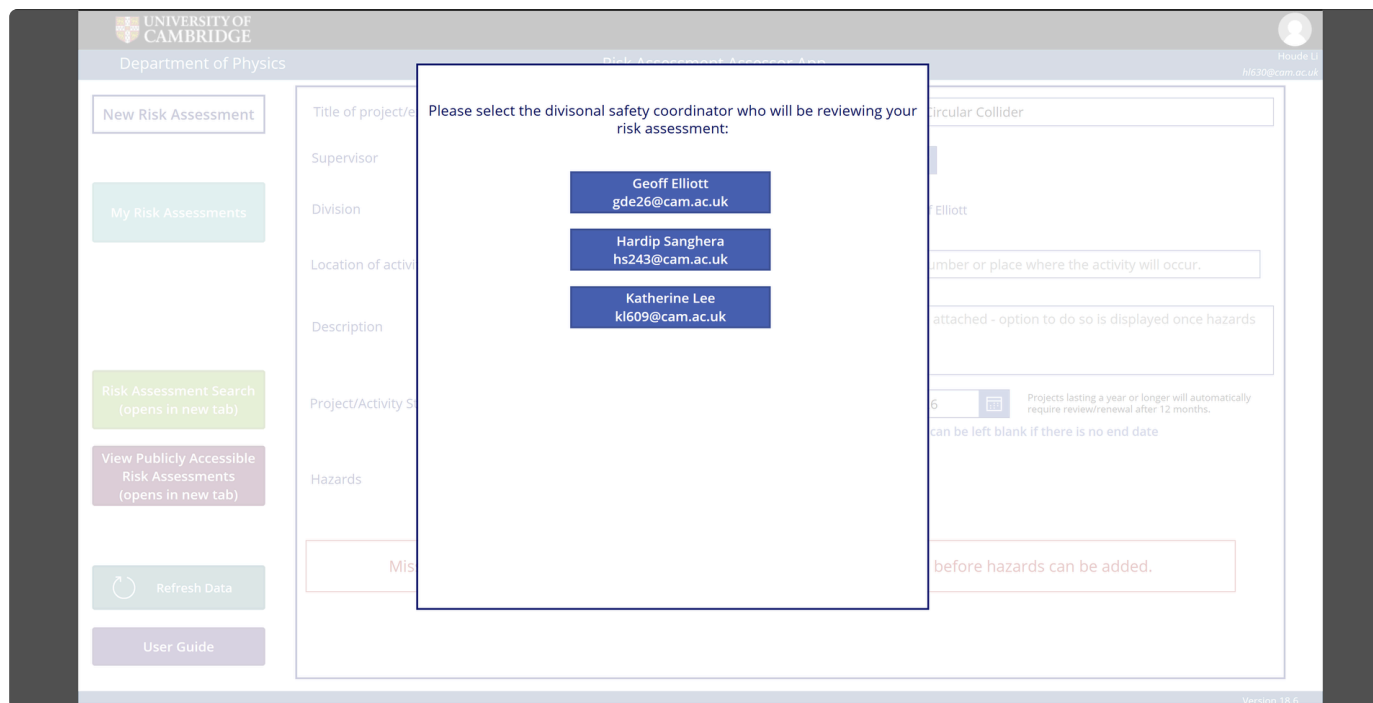
Houde Li  Update your status 09:24

Would we need to a specific room number?

Matthew William Kenzie 09:24

probably could put H&S head of cavendish but you might have to ask louise hirst about that

Houde Li  Update your status 09:24



Would the hazard be "extended computer use"?

(I'll email about the safety coordinator.)

Matthew William Kenzie 09:30

yes

October 26, 2025

Asa Evan Kelsey [15:29](#)

@Houde Li have you figured out what to do / who to select yet?

Houde Li  Update your status [16:24](#)

She hasn't replied yet ...

I'll send another email tomorrow to prompt her. If still no reply, I'll just select one at random.

Asa Evan Kelsey [21:05](#)

Cool thanks

October 27, 2025

Houde Li  Update your status [12:47](#)

No reply. Guess we'll just filling it in then.

Asa Evan Kelsey [13:21](#)

Cool let me know who you choose and I'll match.

Houde Li  Update your status [13:50](#)

I chose Geoff Elliott.

He's the departmental health and safety manager apparently.

Matthew William Kenzie [15:18](#)

I have now signed both of the H&S thingys on the PHY_HS_RiskAssessment app

Houde Li  Update your status [15:46](#)

Eh, Prof. Hirst just replied saying we don't actually need to fill in the risk assessment ...

October 29, 2025

Houde Li  Update your status [15:07](#)

We need the fcc environment for Ixplus, right? I have sent two requests in the last few days and I don't think I have been added yet.

October 30, 2025

Matthew William Kenzie [17:02](#)

do you mean the computing group? it's not essential i don't think

Houde Li  Update your status [17:26](#)

Yeah. I only signed up for the default one.

November 06, 2025

Matthew William Kenzie [22:12](#)

Hi @both. I was thinking that we could make a start on planning out some of our studies. I was hoping you could code up some equations and start making some plots. It would be nice if you could code up the general decay rates for a $B_{(s)}^0$ and $\bar{B}_{(s)}^0$ to a CP-eigenstate f .

These equations have the following form (taken from the slides of my lectures I sent you):

$$\Gamma(B_s^0(t) \rightarrow f) \propto e^{-\Gamma_s t} \left[\cosh\left(\frac{\Delta\Gamma_s t}{2}\right) + A_{\Delta\Gamma} \sinh\left(\frac{\Delta\Gamma_s t}{2}\right) + C \cos(\Delta m_s t) - S \sin(\Delta m_s t) \right],$$

$$\Gamma(\bar{B}_s^0(t) \rightarrow f) \propto e^{-\Gamma_s t} \left[\cosh\left(\frac{\Delta\Gamma_s t}{2}\right) + A_{\Delta\Gamma} \sinh\left(\frac{\Delta\Gamma_s t}{2}\right) - C \cos(\Delta m_s t) + S \sin(\Delta m_s t) \right].$$

where the coefficients encode the CP violation in the process and can be expressed in terms of λ_f

$$\lambda_f = \frac{q}{p} \frac{\bar{A}_f}{A_f}$$

with

$$C = \frac{1 - |\lambda_f|^2}{1 + |\lambda_f|^2}, \quad S = \frac{2 \Im \lambda_f}{1 + |\lambda_f|^2}, \quad A_{\Delta\Gamma} = -\frac{2 \Re \lambda_f}{1 + |\lambda_f|^2}.$$

It would be helpful for you to sit down with these equations and then think about how you could generate some events from these. It would also be nice to derive equations for the CP-asymmetry:

$$A_{CP}(t) = \frac{\bar{\Gamma}(t) - \Gamma(t)}{\bar{\Gamma}(t) + \Gamma(t)} = ?$$

There are simplified cases we can exploit e.g.:

- For a B^0 meson (like in the $B^0 \rightarrow J/\psi K_S^0$ decay) then $\Delta\Gamma \approx 0$
- For both $B^0 \rightarrow J/\psi K_S^0$ and $B_s^0 \rightarrow J/\psi \phi$ there is approximately no CP violation in the decay $|\lambda_f| = 1$ and this then allows one to write $\lambda_f = \eta_f e^{-i\varphi}$ where φ is the CP-violating phase and η_f is the CP-eigenvalue (and you should can then write down what this means for C , S and $A_{\Delta\Gamma}$).
- There is an additional complexity for the $J/\psi \phi$ mode which is not a pure CP-eigenstate but an admixture of CP-even and CP-odd

What I would really like you to do is if I tell you we expect N signal events of a given decay to then simulate what the decay rates and CP asymmetries would look like. I can show you how to do this once you have the equations coded. So a starting point would be to simply make of the decay rates and asymmetries as a function of time based on these kinds of equations

Show less

November 08, 2025

Matthew William Kenzie [07:50](#)

This last line should read "make plots of the decay rates and asymmetries as a function of time"

November 10, 2025

Houde Li  Update your status [09:26](#)

A few questions about the lecture handout on CP violation:

In CP violation in decay, you stated that two amplitudes are required to measure CP violation. I understand this as there should be two different decay processes going from the initial meson to the same final state. To be considered as the "same" final state, should the final state leptons of the two processes be the same as well, or do we just need the same final state meson?

In the coupled meson system, are all M_{ij} and Γ_{ij} real?

In CP violation in mixing, I think some signs seem to be wrong?

On slide 12, if $\Delta\Gamma$ and ΔM are defined as H - L, then they should be equal to $-2\Gamma_{12}$ and $-2M_{12}$.

On slide 13, equation 11, all terms of $g_{\pm}$ should have positive signs. This doesn't affect any of the terms on slide 35.

Matthew William Kenzie [09:56](#)

It does not surprise me there are sign errors

November 12, 2025

Matthew William Kenzie [11:32](#)

This should be the standard reference for those formulas, which should not contain any sign errors. Perhaps better to work from here:

<https://pdg.lbl.gov/2022/reviews/rpp2022-rev-cp-violation.pdf>

in particular eqs 13.17a and 13.17b

Asa Evan Kelsey [15:20](#)

Just worked through the algebra from PDG, fairly sure the only error in what you sent us is that A shouldn't have a minus in it before the real part. Not sure if you've done it and can concur Houde?

Asa Evan Kelsey [15:36](#)

Also, I have plots of general decay rates and asymmetries, and am happy to move on to the N signals part

Houde Li  Update your status [15:43](#)

You mean A_{CP} ? I haven't checked that, but I believe all the decay rates are correct in the lecture slides.

Asa Evan Kelsey [15:46](#)

No, I meant $A_{\Delta\Gamma}$ sorry

And was referring to what was sent in this channel

The lecture slides are fine

Houde Li  Update your status [15:53](#)

Yeah, I concur the sign of $A_{\Delta\Gamma}$ is the only one wrong in the channel.

However, I do believe the signs of $g_{\pm}$ are wrong on slide 13, though it doesn't affect the final results.

Houde Li  Update your status [16:16](#)

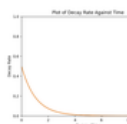
To confirm, I have my plots as well and am also ready to move on.

Matthew William Kenzie [16:36](#)

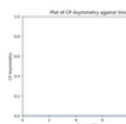
great. please can you share the plots you have?

Houde Li  Update your status [16:47](#)

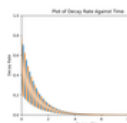
First two with $\lambda_f = 1$, last two with $\lambda_f = 2$. The values of x and y are both for $B_s^0 \leftrightarrow \bar{B}_s^0$.



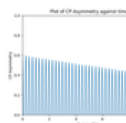
Figure_1.png
PNG 21KB



Figure_2.png
PNG 15KB



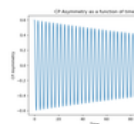
Figure_3.png
PNG 35KB



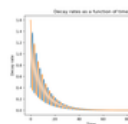
Figure_4.png
PNG 52KB

Asa Evan Kelsey [18:37](#)

Mine look pretty much the same - these are for $\lambda_f = 2\$$



CP_Asymmetry.png
PNG 68KB



Decay rates.png
PNG 39KB

Matthew William Kenzie [20:48](#)

these are for the Bs system I assume, i.e. what values do you use for e.g. $\Delta\Gamma$, Δm

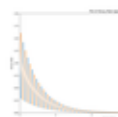
as Asa has it the CP asymmetry can be negative

can we have a look at e.g. cases for where $C=0$ but S and $A_{\Delta\Gamma}$ are not

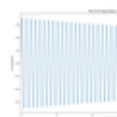
Houde Li Update your status [20:58](#)

I used $x = \frac{\Delta M}{\Gamma} = 25.194$ and $y = \frac{\Delta\Gamma}{2\Gamma} = 0.046$ from the handout.

Here's with $\lambda_f = \frac{1}{\sqrt{2}}(1 + i)$



Decay Rate, 1+1j.p...
PNG 75KB



Asymmetry, 1+1j.png
PNG 156KB

Asa Evan Kelsey [20:59](#)

I just used placeholder values and set $\Delta\Gamma$ to be small; can edit to have proper values

Matthew William Kenzie [21:00](#)

what expressions are you using for A_{CP}

I like the x-scale using Γt and I would suggest can go up to 5 times the lifetime

Asa Evan Kelsey [21:02](#)

I'm using $\frac{\Gamma - \Gamma}{\Gamma - \Gamma}$

plus on the bottom sorry

Houde Li Update your status [21:02](#)

Same

Asa Evan Kelsey [21:05](#)

where in the handout are these?

Houde Li  Update your status [21:05](#)

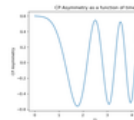
slide 15

Asa Evan Kelsey [21:06](#)

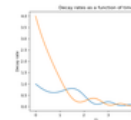
cheers

Asa Evan Kelsey [21:19](#)

my plots are not quite the same as yours now with the given x and y for lambda = 2



CP_Asymmetry (1...
PNG 37KB



Decay rates (1).png
PNG 30KB

Asa Evan Kelsey [21:24](#)

nvm figured it out

November 17, 2025

Houde Li  Update your status [15:09](#)

I seem to have some trouble diagonalising the Hamiltonian $H = M - \frac{1}{2}i\Gamma$. I keep getting the condition $(\frac{q}{p})^2 = \frac{H_{12}}{H_{21}}$, which is the wrong way compared to the one given in the paper [CP Violation in the Quark Sector](#) equation (13.9).

Houde Li  Update your status [15:20](#)

Ah, nevermind, I got the L/H eigenvalues the wrong way ...

Houde Li  Update your status [15:47](#)

Actually, I take that back, I'm still not sure what's happening ...

I'm trying to diagonalise H using a matrix S (the one with p , q , and z) such that the off-diagonal terms in SHS^{-1} vanish, which should constrain p and q . The expression of S as given doesn't constrain the order of L/H eigenvalues since one can exchange them by flipping the signs of q and z .
I attached my work. Would be super grateful if @Asa Evan Kelsey can maybe check it?



diagonalisati...
PDF 513KB

Asa Evan Kelsey [15:56](#)

I could give it a look tomorrow if that's ok? Drowning in TQM supo work slightly.

Houde Li  Update your status [15:56](#)

Sure! Thanks in advance.

November 18, 2025

Asa Evan Kelsey [12:59](#)

Yeah I got the same as you, but PDG seems to be fairly authoritative, so I too am confused

Matthew William Kenzie [15:06](#)

i would not be completely shocked but certainly surprised if there was a mistake in the PDG. let's try and meet before the end of term once you can resolve it (or perhaps not)

btw it's great that the two of you are discussing and working together on this - you should continue to do so on this channel which then acts as your lab book.

November 20, 2025

Houde Li  Update your status [09:42](#)

I think it might be a typo. A later equation on the same page eq 13.12, which makes use of q/p , follows our result correctly as far as I can tell.

November 21, 2025

Asa Evan Kelsey [14:26](#)

Is there anything computational we could be doing next week? Just wary of the progress report due in 2 weeks or so.

Matthew William Kenzie [14:55](#)

yup so the computational thing you can do is the following:

- you now have decay rate equations for B^0 / B^0_{bar}
- so if i give you a sample size number e.g. 100K can you then generate a sample of the decay times of candidates
- if $N=100K$ first throw a random poisson from $N/2$ (assuming equal frac of B^0 and B^0_{bar})
- this you can assume is the number of B and B_{bar} at production ($t=0$)
- then can produce a sample of the decay times for these candidates
- then you can pick some binning scheme in the decay time (say 10 bins from 1 up to 5 average lifetimes or something) - note that normally the bin widths are chosen so that you expect roughly the same number of events in each bin (given these predominantly follow an exponential decay the lower bins are much narrower)
- then you can count the number of events you have (or you can predict with your equations how many you expect in each bin $\int_{t_i}^{t_{i+1}} p(t; \vec{\theta}) dt$)
- you can then compute the time-dependent asymmetry $A_{CP}(t) = (N_{\bar{B}}(t) - N_B(t)) / (N_{\bar{B}}(t) + N_B(t))$ (and you can also provided errors on it because the errors on the number you expect is just the square root of the number.
- you should then overlay the generated points with the asymmetry distribution
- after that you can try fitting the distribution back again

November 25, 2025

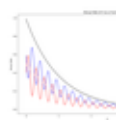
Houde Li  Update your status [13:53](#)

Here's some plots I made so far. Not finalised yet, still has a few things to do.

First plot is decay rates of B . Second plot is decay time distribution for $B \rightarrow f$ with its decay rate (probability distribution function PDF) and cumulative probability distribution function CDF. I used some random parameters for $\lambda_f, \bar{\lambda}_{\bar{f}}, |\frac{q}{p}|$.

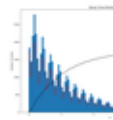
For the sampling, I integrated the decay rate to get the CDF, got a numerical inverse CDF with `scipy.interpolate.interp1d`, then fed a sample drawn from a uniform distribution to the inverse CDF function to give a sample of decay time.

There's another method with `scipy.stats.rv_continuous` that give a sample by analytically solving for the inverse CDF for every candidate of the uniform distribution. I think this method will be much slower with 100k samples.



Decay Rate as a F...

PNG 75KB



Decay Time Distribution...

PNG 68KB

Also, here is the requirement of the progress report on TIS.

I guess we should start writing it about now? Would we need to discuss about the project plan?

6.2. Progress report: Students will be asked to complete a progress report at the end of Michaelmas Term. This initial Report (between 4 and 6 A4 pages in length) should describe the project in their own words, putting the physics into context (including references to the relevant literature) and describing the goals of the project; it must also include a project plan. This report should be electronically signed by both the student and the supervisor to indicate their agreement with the plan and should be uploaded by the student onto the TIS by **Friday 5th December 2025**. Failure to submit an Initial Report will result in the loss of 5% of the available project marks.

November 26, 2025

Matthew William Kenzie [08:45](#)

Yes. Please start writing. Let's try to meet to discuss this week or early next.

I am basically free tomorrow and Friday apart from a lecture 10-11 on Friday

Asa Evan Kelsey [11:53](#)

Should the initial report be in latex as well? As for meetings Friday works best for me.

Matthew William Kenzie [12:00](#)

Can we meet Friday afternoon. Should ideally be latex yeah.

Asa Evan Kelsey [12:12](#)

Works for me, I'm free all afternoon

Houde Li  Update your status [13:55](#)

I have a supervision from 2pm to 3pm on Friday. Can we do 1pm?

Matthew William Kenzie [16:02](#)

yup 1pm on Friday is good for me

Matthew William Kenzie [16:06](#)

it's nice that you figure how to generated from this directly (i.e using the inverse of the cdf). a couple of things i notice is that you have the rates for $X \rightarrow f$ and $X \rightarrow \bar{f}$ but in these analyses the final state f is CP symmetry (it's $J/\psi K_S^0$ or $J/\psi \phi$) but what we can do is tag the B-meson flavour when it's produced, i.e. at $t = 0$ so what we want to know is the distribution $X \rightarrow f$ and $\bar{X} \rightarrow f$ - perhaps this is the same for CP symmetric f - so then just a plot labelling issue.

the toys you have are nice but there seems to be an issue with the normalisation.

would be great to see this histogram for mesons initially (i.e. at $t=0$) produced as B^0 or B^0_{bar} (as a blue his and an orange hist say)

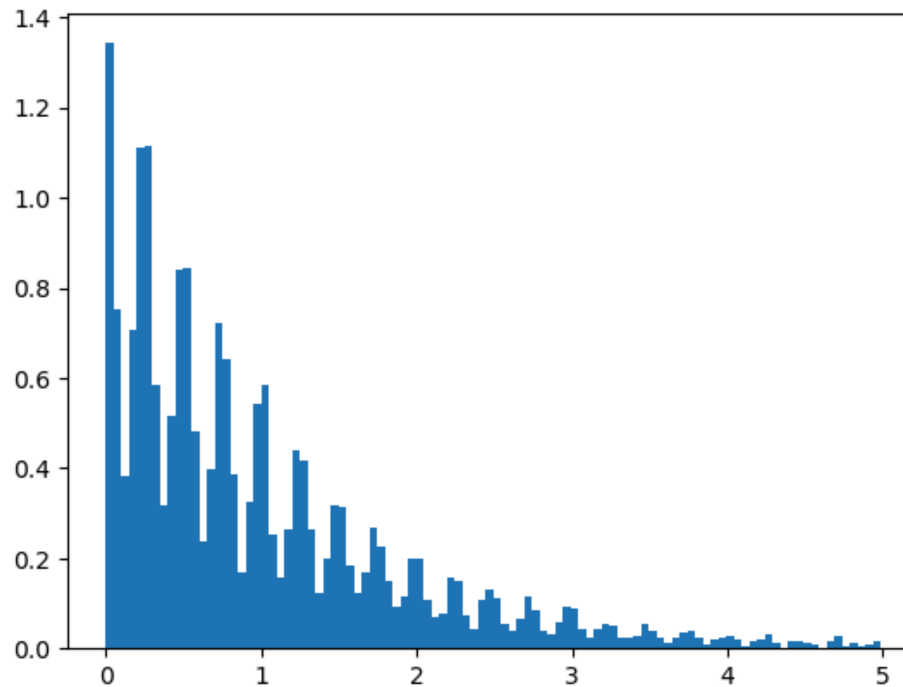
Matthew William Kenzie [16:07](#)

see you at 1pm on Friday in my office - would be helpful if you could also sketch out before hand (just in terms of section / subsection headings) what your progress report will look like. it's 4-6 pages and you should save probably 2 pages for the plan - which we can discuss in the meeting

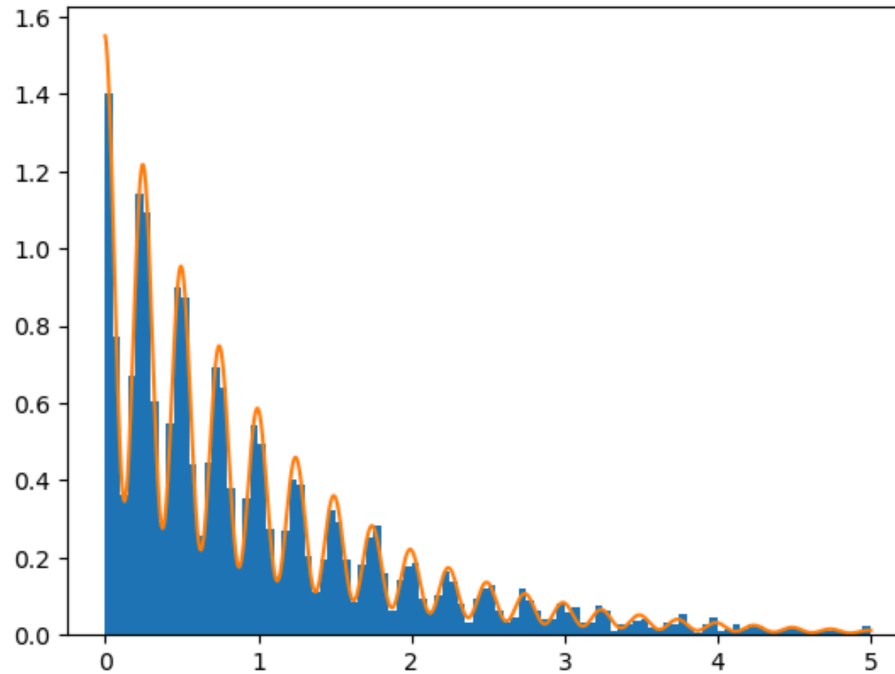
November 27, 2025

Asa Evan Kelsey [15:08](#)

wondering if the normalisation on this is off on account of it going above 1? this is sampling from the inverse cdf, and I'm pretty sure I normalised the CDF and PDF right



looks like this with the pdf overlayed



Houde Li  Update your status 15:14

I think the PDF is the derivative of the CDF, so as long as the CDF is one at $t \rightarrow +\infty$ it's fine.



Matthew William Kenzie 15:26

looks pretty good to me.

November 28, 2025

Houde Li  Update your status 13:00

(In lecture at the moment, will be a few minutes late.)

Matthew William Kenzie 13:28

<https://scikit-hep.org/iminuit/>

https://scikit-hep.org/iminuit/notebooks/cost_functions.html#Unbinned-fit

https://scikit-hep.org/iminuit/notebooks/cost_functions.html#Extended-unbinned-fit

```
def pdf_Bz(t, C, S, D):  
  
def generate(q, C, S, D, N=25K):  
  
t_i_Bz = generate(q=1, C, S, D, N=25K)  
t_i_Bzb = generate(q=1, C, S, D, N=25K)  
  
from iminuit import cost, Minuit  
  
n2ll = cost.UnbinnedNLL(t_i_Bz, pdf_Bz) # define cost function to minimize  
  
mi = Minuit(n2ll, C=1, S=1, D=0) # make minimiser  
mi.migrad() # run minimisation  
mi.hesse() # get error matrix
```

```

def pdf_Bz(t, C, S, D):

def pdf_Bzb(t, C, S):

def density(t, Nb, Nbbbar, C, S, D):
    return Nb + Nbbbar, Nb*pdf_Bz(t, C, S, D) + Nbbbar*pdf_Bzb(t, C, S, D)

def generate(q, C, S, D, N=25K):

N = 50K
N_Bz = np.random.poisson( N / 2)
N_Bzb = np.random.poisson( N / 2)

t_i_Bz = generate(q=1, C, S, D, N=N_Bz)
t_i_Bzb = generate(q=1, C, S, D, N=N_Bzb)

t_i = np.concatenate([t_i_Bz, t_i_Bzb])

from iminuit import cost, Minuit

n2ll = cost.ExtendedUnbinnedNLL(t_i, density) # define cost function to minimize

mi = Minuit(n2ll, Nb=50K/2, N_Bzb=50K/2, C=1, S=1, D=0) # make minimiser
mi.limits["Nb", "Nbbbar"] = (0, 50K)
mi.limits["C", "S", "D"] = (-1, 1)
mi.migrad() # run minimisation
mi.hesse()  # get error matrix

```

```
mi.limits
```

Houde Li  Update your status 15:16

In the case when X^0 and $\bar{X}^0$ decay to a CP eigenstate, which is the case for B^0 , I found that the two normalisation conditions for the two decay channels are actually independent. One would give an expression for $|A_f|^2$ and the other would give something like this:

$$\frac{1 + \left(\frac{\Delta M}{\Gamma}\right)^2}{1 - \left(\frac{\Delta \Gamma}{2\Gamma}\right)^2} \frac{1 + |\lambda_f|^2 + \frac{\Delta \Gamma}{2\Gamma} 2\text{Re}\{\lambda_f\}}{1 - |\lambda_f|^2 - \frac{\Delta M}{\Gamma} 2\text{Im}\{\lambda_f\}} = \frac{1 + \left|\frac{q}{p}\right|^2}{1 - \left|\frac{q}{p}\right|^2}$$

Since the decay rates are only dependent on λ_f and $\left|\frac{q}{p}\right|^2$ (assuming ΔM , $\Delta \Gamma$, and Γ are known), this condition would reduce the degrees of freedom to only 2.

EDIT: The above is really just saying normalisation fixes $|A_f|$ and $|\bar{A}_f|$, leaving $\arg\left(\frac{A_f}{\bar{A}_f}\right)$ as the only degree of freedom in decay; then we also have another degree of freedom $\left|\frac{q}{p}\right|$ from mixing.

Not sure if this will help with the fitting since the dependence is not simple, I just think this might be worth mentioning.

Matthew William Kenzie 16:48

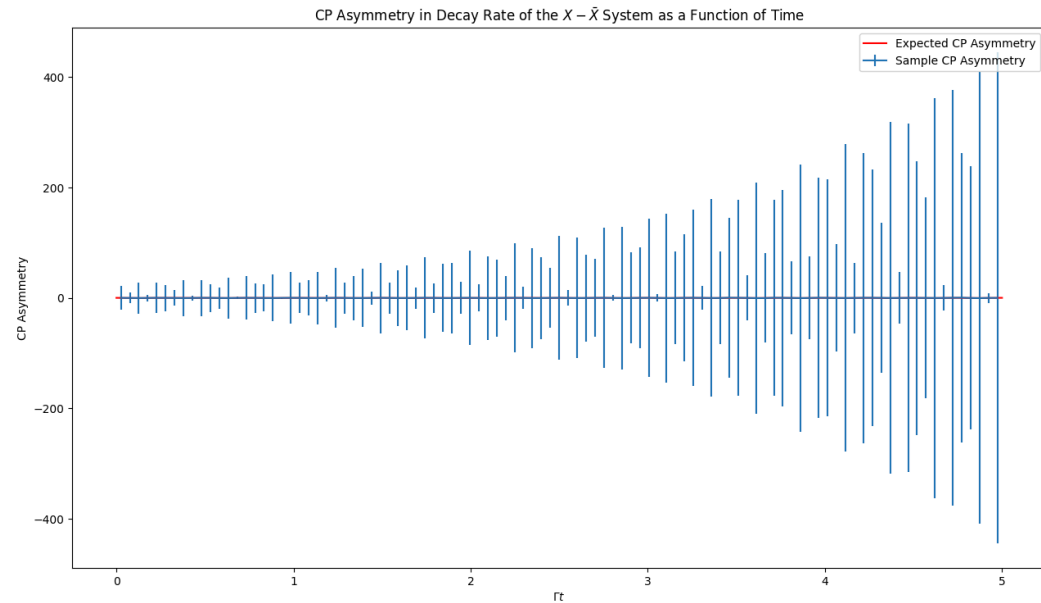
that's a very good point

and I guess it helpful in environment when you know exactly how many of each flavour are produced (most cases we don't)

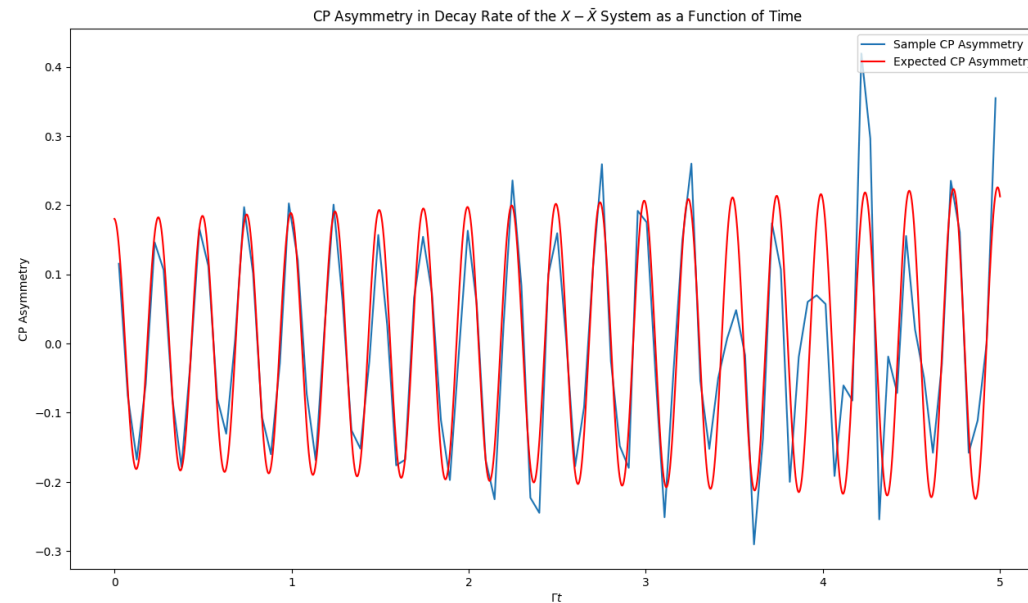
November 30, 2025

Houde Li  Update your status 17:30

Just plotted the CP asymmetry. I tried to account for both the poisson uncertainty in N and the uncertainty due to sampling. The errorbars look massive ...



It looks okay without the errorbars.



Also, I assume the N_s in the definition in A_{CP} refer to the number of decays in a bin at a time t instead of the number of remaining mesons, right?

Houde Li  Update your status 17:46

If I remove the uncertainty from sampling (I modelled as square root of binomial variance), I get essentially 0 errors, which can't account for the fluctuation in the plot.

Matthew William Kenzie 22:04

if the errors are too large then the binning is too fine

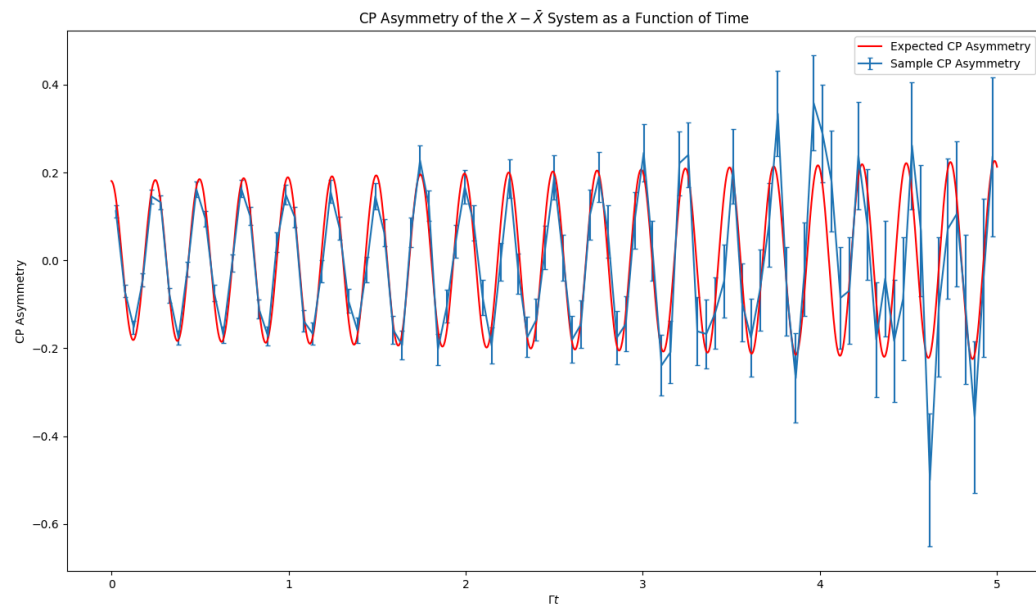
~~if the errors are too large then the binning is too fine~~
i think the error on the asymmetry should come out as:

$$\sigma = \sqrt{\frac{1 - A^2}{N_{\bar{B}} + N_B}}$$

December 01, 2025

Houde Li  Update your status 09:03

I got the expression of the error wrong. It looks much better now. Thanks!

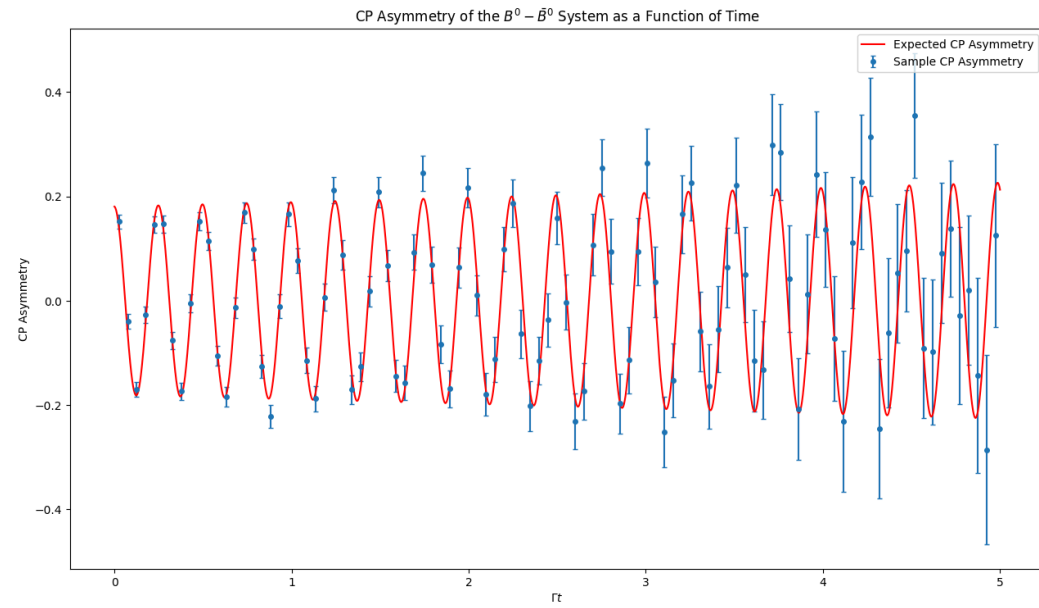


Matthew William Kenzie 10:48

This looks good. Can you draw the points without a connecting line:

```
ax.errorbar(x, y, yerr, fmt=ko)
```

Houde Li

 Update your status11:19

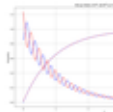
Houde Li  Update your status 11:25

Also updated the other plots a bit.

For the decay rates plot, I tried to do a interchanging dash line for the CDFs.

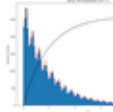
For the decay time distributions, I added a seperate bin for the events beyond $\Gamma t = 5$.

I hope this is okay. Are the colours a bit jarring?



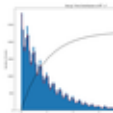
Decay Rates.png

PNG 74KB



Decay Time Distribution...

PNG 65KB



Decay Time Distribution...

PNG 67KB

Matthew William Kenzie 12:38

these are brilliant. you can chuck the overflow away

so how many events are in your sample for that plot?

- question now is can you add a dashed green line to the plot which shows the distribution fitted to the sample (add to legend as "Measured CP Asymmetry") and can you print or display the `iminuit.Minuit` object after the minimisation has run (`mi.migrad()`, `mi.hesse()`). this should show the fitted value of S , C , D with errors and covariance, accesible as:

```
mi.values
mi.errors
mi.covariance
```

hopefully your sample contains some Poisson variant number on an expectation (what is that value?) and then when you fit back you fit it extended so that you get fitted number of B and $Bbar$

could you also add to your plots what the generated values of S , C and D are

Houde Li  Update your status 12:55

The initial numbers of B and $\bar{B}$ were both drawn from $\text{Poisson}(0.5 \cdot 100k)$.

I haven't looked too much into iminuit yet. I'll do that and write the report in the next two days hopefully.

Thanks for the feedback!

Asa Evan Kelsey [15:39](#)

in the report due friday, how far would you say we should get to in the coding i.e. should I put everything that Houde has put in here in my report and if so should I also do the iminuit stuff

Asa Evan Kelsey [15:43](#)

What is A in this

Asa Evan Kelsey [15:44](#)

oh nevermind is it just the A_CP

December 02, 2025

Matthew William Kenzie [12:43](#)

don't worry too much about getting into the coding. give an overview of the formulas and where they come from. then show some of these plots which give you distributions, asymmetries. show the distrbutions of the toys you have (like the plots Houde has sent).

Don't worry too much about getting the iminuit stuff into the progress repotr if you don't have time.

Send the prog report when you have it and I'll sign it

Asa Evan Kelsey [12:56](#)

have sent over on mattermost dm

Matthew William Kenzie [14:25](#)

thanks for the report Asa. the intro stuff is very nice.

- Fig 4 and Fig 5 have different captions but look similar (does there need to be some replacement here?)

for the plan - i think you could add some more details on the following steps:

- For early Feb meeting - ability to fit back generated samples for CP parameters
- Feb - Mar work on analysing simulated $B^0 \rightarrow J/\psi K^0$ and $B_s^0 \rightarrow J/\psi \phi$ decays at the IDEA detector of FCC-ee running at the Z threshold. this work will determine approximately the number of signal decay for each mode one can expect and important what the background contributions look like (are they easy or difficult to distinguish from the signal) and how much background is there. the final thing to determine from this simulation is what the decay-time-acceptance will look like once the selection is in place.
- Mar - Apr. incorporate the signal/ background ratio and then time acceptance into the sampling from your p.d.f. so you can then simulate the sensitivity to C, S, D under different scenarios
- Mar - Apr. also try to incorporate the flavour tagging aspect (real data scenario does not tag with perfect efficiency) so would need to incorporate the mis-tag rate
- possible extensions. set up something similar for $B^0 \rightarrow D^0 \pi^0$, $B_s \rightarrow D^0 K^0$

Asa Evan Kelsey [14:29](#)

I think the similarity of the figs may be due to their size - will scale up. Will also incorporate the dates for the plan

Matthew William Kenzie [14:29](#)

no but the caption on the left say asymmetry but it doesn't look like an asymmetry to me

Asa Evan Kelsey [14:29](#)

ah yes oops slight mistake

December 03, 2025

Houde Li  Update your status [11:08](#)

I dm'ed the report as well. Didn't have time to do iminuit.

December 04, 2025

Asa Evan Kelsey [18:33](#)

Hi Matt, do you know what time tomorrow they have to be signed and submitted? My broadband is currently down for the count so need to work around it.

Houde Li  Update your status 18:43

It says 4pm on TIS.

Matthew William Kenzie 22:34

signed version returned to you

i'm afraid i don't know about deadlines

January 23

Houde Li  Update your status 12:04

I'm done with my exams this term. I'll send a summary of what I've done over Christmas before next Monday.

January 25

Houde Li  Update your status 11:52



vacation_summar...
PDF 2.6MB

Houde Li  Update your status 12:21

Also, the lab notes submission only accepts a url. Can the admin of the chat group export the chat history from mattermost? This way we can paste it to onenote etc to get a url.

January 27

Matthew William Kenzie 13:01

I'm not a system admin so can't export the chat annoyingly. When do you need the url for because we could either make the channel public or I could ask the system admin to export the chat for me

thanks for the updates **Houde**. Could I ask you to make some plots of the pull distributions on CDS and also λ_f under your framework. the pull is defined as the fitted value minus the generated value divided by the fitted error. also good to make histograms of the fitted values and then put on a dashed line where the generated value is for each parameter. in an ideal world the pull distribution is a unit Gaussian. putting sensible limits on the parameters will often help the minimiser (and can help it move away from areas which result in a negative pdf).

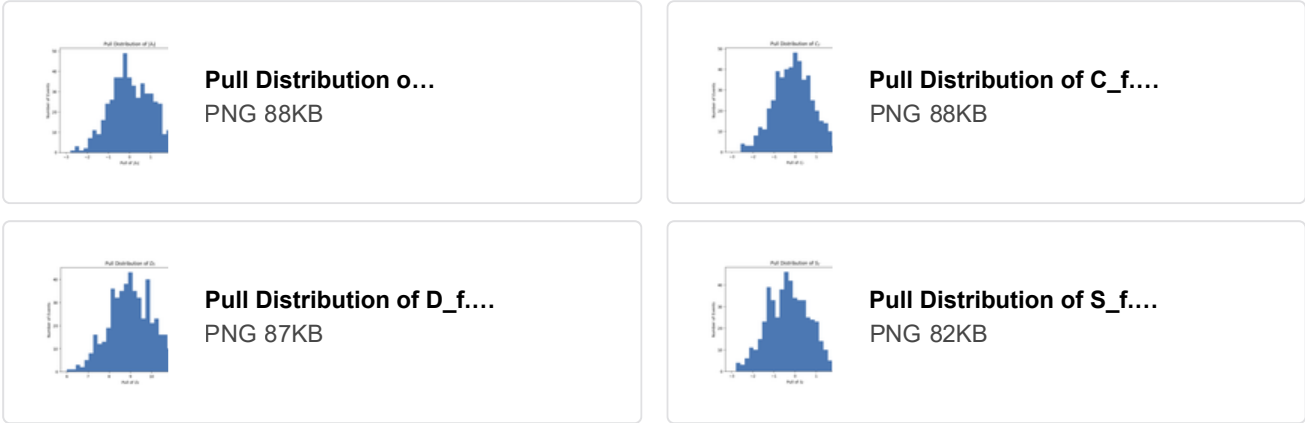
as exams are over i think it's probably time for us to have a meet up to discuss next steps

i would also like you to come along to my group meetings.

in particular for the next two weeks our two 1st year PhD students will give talks on their progress which would be good for you to see
the group meetings are on Tuesdays at 10:00
usually in RDC B2 but i don't know the room for next week yet

Houde Li  Update your status 13:09
Notebook submission deadline is June 1st.

Houde Li  Update your status 22:28
Pull distributions for fitting $|\lambda_f|$ and for fitting CDS (still with $-1 < C, D, S < +1$ box limit).



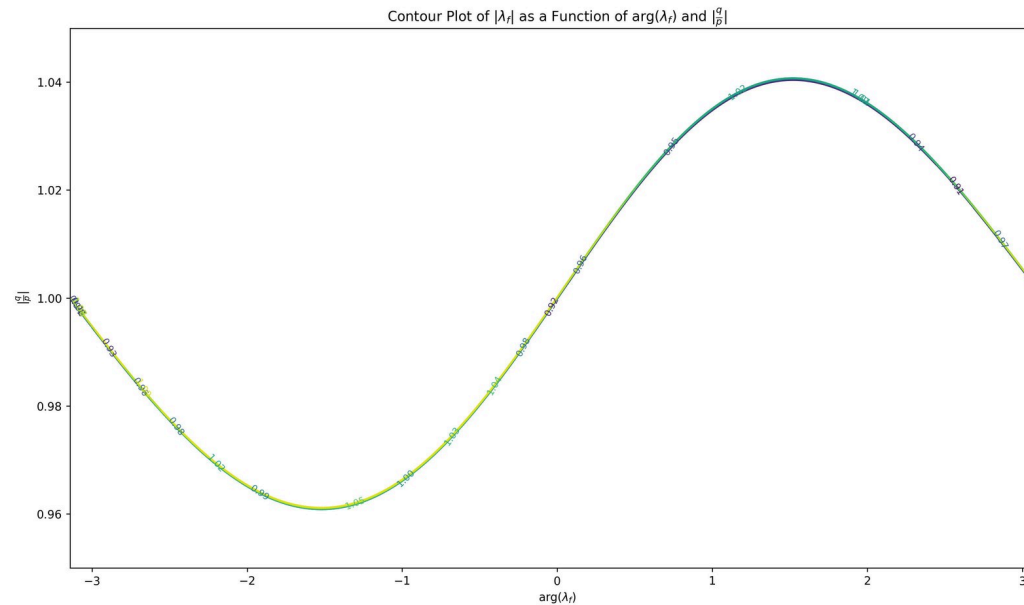
January 28

Houde Li  Update your status 15:36
Found the (other) problem with my fitting. The samples used for fitting were generated with the same code as for when I was making the PDF plots, which only had 5 lifetimes. Fitting CDS works for me now.

January 29

Houde Li  Update your status 10:08

Plotted $|\lambda_f|$ as a function of $\arg(\lambda_f)$ and $|\frac{q}{p}|$, showing only $0.9 < |\lambda_f| < 1.1$. It would seem that knowing $|\frac{q}{p}|$ would immediately give $\arg(\lambda_f)$ up to a rather high precision provided we assume $|\lambda_f|$ is anywhere close to 1.



January 30

Matthew William Kenzie 15:25

thanks for the updates **Houde**. Are you guys available to meet next week?

Houde Li  Update your status 15:27

I'm fine with that.

Asa Evan Kelsey 21:30

Yeah that works for me. My mornings are pretty universally clear

February 06

Matthew William Kenzie 10:20

<https://uproot.readthedocs.io/en/latest/>

Matthew William Kenzie 10:25

<https://pdg.lbl.gov/2024/reviews/rpp2024-rev-monte-carlo-numbering.pdf>

Matthew William Kenzie 10:54

```
source /cvmfs/sw.hsf.org/key4hep/setup.sh
```

```
/eos/user/m/mkenzie/fcc/
```

February 08

Matthew William Kenzie 14:37

files are now copied over to `/eos/user/m/mkenzie/fcc/kenzie_CPV_tuples_no_selection`

here is the hack of the script i was putting together to do what i asked:

```

import uproot
import numpy as np
import awkward as ak
import vector
vector.register_awkward()

from branches import MC_branches, Rec_branches

# --- PDG masses in GeV ---
M_JPSI = 3.096900
M_PHI  = 1.019461

# --- mass windows (tune as you like) ---
JPSI_WIN = 0.15 # +/- 150 MeV
PHI_WIN  = 0.02 # +/- 20 MeV

tr = uproot.open("kenzie_CPV_tuples_no_selection/p8_ee_Zbb_ecm91_EvtGen_Bs2JpsiPhiSVV/chunk_0.root")

tr.show()

# MC_parts = tr.arrays(MC_branches)

events = tr.arrays(Rec_branches, entry_stop=10_000)

# 1) zip per-particle fields into a particle record array: events -> list[particles]
parts = ak.zip(
    {
        "pdg": ak.values_astype(np rint(events["Rec_true_PDG"]), "int32"),
        "px":  events["Rec_px"],
        "py":  events["Rec_py"],
        "pz":  events["Rec_pz"],
        "E":   events["Rec_e"],
        "q":   events["Rec_q"],
    },
    depth_limit=2,
)

# 2) select by PDG sign
mu_plus = parts[parts.pdg == -13]
mu_minus = parts[parts.pdg == 13]

k_plus = parts[parts.pdg == 321]
k_minus = parts[parts.pdg == -321]

# 3) all opposite-sign pairs within each event
mu_pairs = ak.cartesian({"mup": mu_plus, "mum": mu_minus}, axis=1, nested=True)

```

```

k_pairs = ak.cartesian({"kp": k_plus, "km": k_minus }, axis=1, nested=True)

# 4) require at least one of each in the event (optional but common)
has_mu_pair = ak.num(mu_pairs.mup, axis=1) > 0
has_k_pair = ak.num(k_pairs.kp, axis=1) > 0
good_events = has_mu_pair & has_k_pair

# 5) apply good events selection

# 5) build 4-vectors
mup4 = vector.zip({"px": mu_pairs.mup.px, "py": mu_pairs.mup.py, "pz": mu_pairs.mup.pz, "E": mu_pairs.mup.E})
mum4 = vector.zip({"px": mu_pairs.mum.px, "py": mu_pairs.mum.py, "pz": mu_pairs.mum.pz, "E": mu_pairs.mum.E})
jpsi4 = mup4 + mum4
m_jpsi = jpsi4.mass

kp4 = vector.zip({"px": k_pairs.kp.px, "py": k_pairs.kp.py, "pz": k_pairs.kp.pz, "E": k_pairs.kp.E})
km4 = vector.zip({"px": k_pairs.km.px, "py": k_pairs.km.py, "pz": k_pairs.km.pz, "E": k_pairs.km.E})
phi4 = kp4 + km4
m_phi = phi4.mass

# 1) Candidate-level mass selections
jpsi_sel = (m_jpsi > (M_JPSI - JPSI_WIN)) & (m_jpsi < (M_JPSI + JPSI_WIN))
phi_sel = (m_phi > (M_PHI - PHI_WIN)) & (m_phi < (M_PHI + PHI_WIN))

jpsi4_sel = jpsi4[jpsi_sel]
m_jpsi_sel = m_jpsi[jpsi_sel]

phi4_sel = phi4[phi_sel]
m_phi_sel = m_phi[phi_sel]

# 2) Keep only events that still have >=1 selected candidate of each type
has_jpsi = ak.num(jpsi4_sel, axis=1) > 0
has_phi = ak.num(phi4_sel, axis=1) > 0
evt_sel = has_jpsi & has_phi

jpsi4_sel = jpsi4_sel[evt_sel]
m_jpsi_sel = m_jpsi_sel[evt_sel]

phi4_sel = phi4_sel[evt_sel]
m_phi_sel = m_phi_sel[evt_sel]

# 3) Build all B candidates per event: cartesian product of selected J/psi and phi
# B_cands = ak.cartesian({"jpsi": jpsi4_sel, "phi": phi4_sel}, axis=1, nested=True)
# B4 = B_cands.jpsi + B_cands.phi
# m_B = B4.mass

```

```
# 4) OPTIONAL (recommended): choose ONE best B candidate per event
# Here: choose the combination minimizing (|mJ/psi - PDG| + |mphi - PDG|)
# mJ_in_combo = B_cands.jpsi.mass
# mP_in_combo = B_cands.phi.mass
# score = ak.abs(mJ_in_combo - M_JPSI) + ak.abs(mP_in_combo - M_PHI)
#
# best_idx = ak.argmin(score, axis=1)
# best_jpsi4 = B_cands.jpsi[best_idx]
# best_phi4 = B_cands.phi[best_idx]
# best_B4 = best_jpsi4 + best_phi4
#
# best_m_jpsi = best_jpsi4.mass
# best_m_phi = best_phi4.mass
# best_m_B = best_B4.mass
```

[Show less](#)

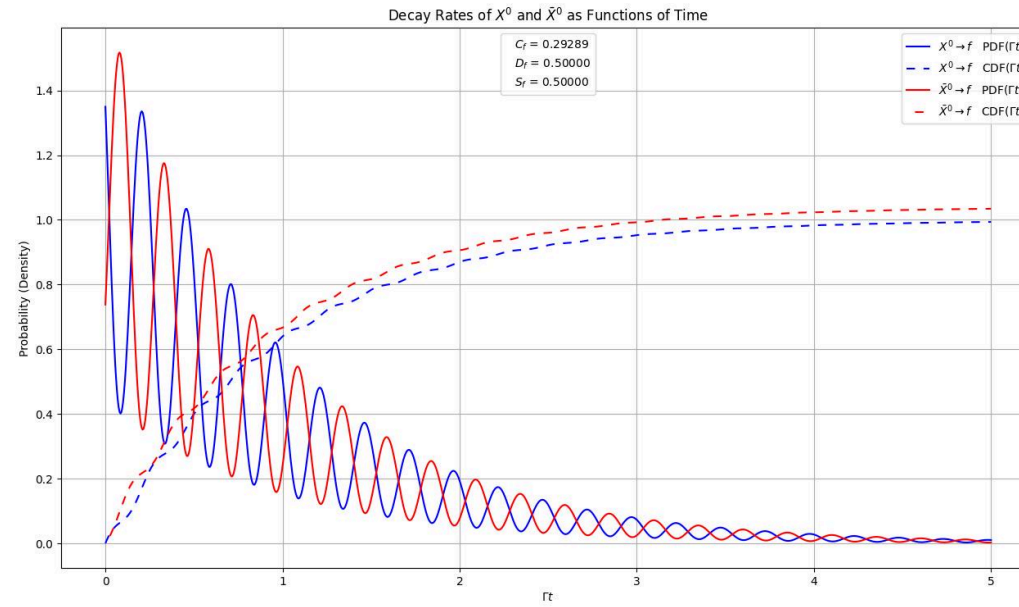
February 16

Houde Li  Update your status 10:05

Got around to fix some bugs with the fitting. The decay rates' normalisations are now proportional to their branching fractions as predicted by CDS. Results from the fitting are plotted.

Matthew William Kenzie 10:06**Houde Li**  Update your status 10:06

Decay rates:



Summary of fitting:

True parameter values

$C_f = 0.2928932188134524$

$D_f = 0.5$

$S_f = 0.5$

IMINUIT fitting results over 100 runs, each with 49019 events

$C_f = 0.29262 \pm 0.00576 \text{ (sd)} \pm 0.00058 \text{ (se)}$

$D_f = 0.50438 \pm 0.06880 \text{ (sd)} \pm 0.00733 \text{ (se)}$

$S_f = 0.49993 \pm 0.00537 \text{ (sd)} \pm 0.00056 \text{ (se)}$

$C_{f_pull} = -0.04791 \pm 0.99157 \text{ (sd)}$

$D_{f_pull} = 0.04668 \pm 0.94370 \text{ (sd)}$

$S_{f_pull} = -0.01406 \pm 0.95210 \text{ (sd)}$

Gradients of the unbinned cost function at the true and fitted parameter values

$grad_iminuit_C_f = 0.64916 \pm 0.86647 \text{ (sd)}$

$grad_iminuit_D_f = -0.03778 \pm 0.11869 \text{ (sd)}$

$grad_iminuit_S_f = -1.02804 \pm 0.70118 \text{ (sd)}$

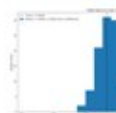
$grad_true_C_f = 15.47048 \pm 362.66652 \text{ (sd)}$

$grad_true_D_f = -0.72859 \pm 28.96221 \text{ (sd)}$

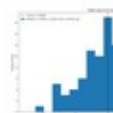
$grad_true_S_f = 1.72535 \pm 363.86801 \text{ (sd)}$

where, for the fitted parameter results, "sd" is the sample standard deviation and "se" is the standard error. They are quite different, so I quoted both.

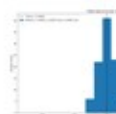
Histograms of fitted values of the CDS parameters:



Fit_C.jpeg
JPEG 47KB

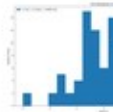


Fit_D.jpeg
JPEG 48KB

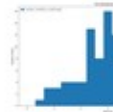


Fit_S.jpeg
JPEG 47KB

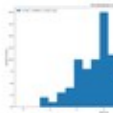
Pull distributions of CDS:



Pull_C.jpeg
JPEG 46KB



Pull_D.jpeg
JPEG 43KB



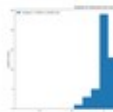
Pull_S.jpeg
JPEG 44KB

Also looked at the gradients of the unbinned cost function at the true and fitted parameter values. At the fitted values, the gradients should be zero for a good/working fit. At the true values, the expectation should be zero, and the variance is expected to scale with the sample size.

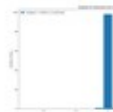
Matthew William Kenzie [10:12](#)

these are very nice. so when you have the distribution of pulls that you have. can you quote the mean of the pull and it's error (std mean on error) and then the width of the pull (standard deviation of the pull) and its error (which is $\sigma/\sqrt{2N+1}$) so that you have $\mu = \text{mean} \pm \text{std error on mean}$, and $\sigma = \text{standard deviation} \pm \text{std error on std dev}$

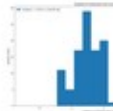
Houde Li  Update your status [10:12](#)



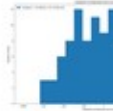
Grad_Fit_C.jpeg
JPEG 45KB



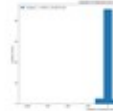
Grad_Fit_D.jpeg
JPEG 42KB



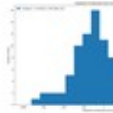
Grad_Fit_S.jpeg
JPEG 44KB



Grad_True_C.jpeg
JPEG 50KB



Grad_True_D.jpeg
JPEG 44KB



Grad_True_S.jpeg
JPEG 48KB

Right. I forgot to do error propagation for the pull. Will add that soon.

February 17

Houde Li  Update your status [13:10](#)

Defining the pull of each fitting as $\frac{x_i - x_0}{\sigma_i}$, where $x_i \pm \sigma_i$ is the result of one fitting and x_0 is the true value, I think the errors in x_i and σ_i are needed to calculate the error on the mean/width for the pull distribution. Should I take those to be the error in the mean/width of the fitted parameter value?

Edit: only the error on the mean for the pull needs the errors in x_i and σ_i

Houde Li  Update your status 13:24

I did that and the result looks like this.

True parameter values

$C_f = 0.2928932188134524$

$D_f = 0.5$

$S_f = 0.5$

IMINUIT fitting results for over 100 runs, each with 49152 events

C_f : mean = 0.29354 ± 0.00058 , width = 0.00647 ± 0.00046

D_f : mean = 0.49555 ± 0.00733 , width = 0.07302 ± 0.00519

S_f : mean = 0.50008 ± 0.00056 , width = 0.00568 ± 0.00040

C_f _pull: mean = 0.11042 ± 0.01000 , width = 1.11323 ± 0.07911

D_f _pull: mean = -0.10486 ± 0.01300 , width = 0.99422 ± 0.07066

S_f _pull: mean = 0.01094 ± 0.01011 , width = 1.01011 ± 0.07179

Matthew William Kenzie [13:25](#)

no you don't need that and you don't need to propagate. let's say you ran N toys (i.e. N generations), where $N = 250$ is usually considered a good number ($N = 100$ can suffice if it's particular compute expensive). then you have a sample of $N = 250$ values of the pull, $p_i = \frac{x_i - x_0}{\sigma_i}$. you can compute sample estimates of the mean and standard deviation of this sample according to the standard formula (as well as sample estimates of the errors on those estimates, often called the "standard error on the mean" and the "standard error on the standard deviation"):

Mean:

$$\hat{\mu} = \frac{1}{N} \sum_i p_i$$

```
np.mean(p_i)
```

Width (standard deviation):

$$\hat{\sigma} = \sqrt{\frac{1}{N-1} \sum_i (p_i - \hat{\mu})^2}$$

```
np.std(p_i, ddof=1)
```

(note here that the `ddof=1` i.e. $1/(N-1)$ factor is Bessel's correction which accounts for the bias in that the same sample is also used to estimate the mean $\hat{\mu}$)

Standard Error on the mean:

$$\hat{\sigma}_{\hat{\mu}} = \frac{\hat{\sigma}}{\sqrt{N}}$$

```
np.std(p_i, ddof=1) / N**0.5
```

Standard Error on the standard deviation:

$$\hat{\sigma}_{\hat{\sigma}} = \frac{\hat{\sigma}}{\sqrt{2(N-1)}}$$

```
np.std(p_i, ddof=1) / (2*(N-1))**0.5
```

[Show less](#)

then what i want you to quote is $\hat{\mu} \pm \hat{\sigma}_{\hat{\mu}}$ (which should be consistent with 0) and $\hat{\sigma} \pm \hat{\sigma}_{\hat{\sigma}}$ (which should be consistent with 1)

Houde Li  Update your status [13:28](#)

Got it. Thank!

Matthew William Kenzie [13:29](#)

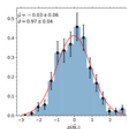
you can also make what i call the "Aidan plot"

see Sec 3.11.3 (parametric bootstrapping) and Sec 3.1.3 (samples estimates of mean, variance and standard deviation) in my stats lecture notes:

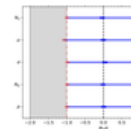


StatisticalMethods...
PDF 4.9MB

so for example you can try and make your histogram plots look like this (and notice you don't need both the filled histograms and the error bar points) - probably just the errorbar points. and then summarise them with the Aidan plot (the one below) which should look consistent with 0 and width 1



Screenshot 2026-0...
PNG 53KB



Screenshot 2026-02-17 ...
PNG 125KB

Houde Li  Update your status [13:36](#)

Changed my coded and reran it:

True parameter values

$C_f = 0.2928932188134524$

$D_f = 0.5$

$S_f = 0.5$

IMINUIT fitting results over 250 runs, each with 49267 events

C_f : mean = 0.29370 ± 0.00037 , width = 0.00608 ± 0.00027

D_f : mean = 0.49925 ± 0.00462 , width = 0.07076 ± 0.00317

S_f : mean = 0.49984 ± 0.00036 , width = 0.00568 ± 0.00025

C_f _pull: mean = 0.13972 ± 0.06618 , width = 1.04644 ± 0.04689

D_f _pull: mean = -0.02420 ± 0.06100 , width = 0.96444 ± 0.04322

S_f _pull: mean = -0.03020 ± 0.06389 , width = 1.01017 ± 0.04527

Houde Li  Update your status [13:42](#)

Do you know if we'll need to apply for more computation resources for the project?

I haven't looked at the uproot stuff just yet.

Matthew William Kenzie [14:11](#)

yes quite possibly

please make a request as per Ulrich's email

is there any chance you guys are free to meet tomorrow afternoon instead - I cannot do Friday anymore

Houde Li  Update your status [14:17](#)

I can do that.

Matthew William Kenzie [14:17](#)

tomorrow at either 2pm or 3pm?

Houde Li  Update your status [14:17](#)

Both are fine for me.

[@Asa Evan Kelsey](#)

Asa Evan Kelsey [14:35](#)

I can do 2pm

Matthew William Kenzie [15:20](#)

ok let's do 2pm tomorrow (i may be 5-10 mins late)

February 19

Asa Evan Kelsey [10:41](#)

[@Houde Li](#) Just saw the mid range compute request, do I not need to sign up independently?

Houde Li  Update your status [10:42](#)

Let me check.

I think you need to submit one as well. [@Asa Evan Kelsey](#)

Asa Evan Kelsey [10:49](#)

Cool thanks - as for any specific software stuff what did you put? And also which OS?

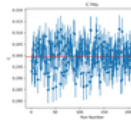
Houde Li  Update your status [10:51](#)

I think you can see my ticket, right? I just put windows and python.

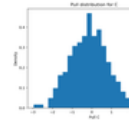
February 23

Asa Evan Kelsey [15:43](#)

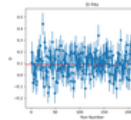
realise I haven't put much in here recently so just gonna dump this in here for the lab notebook: these are all for 250 runs of iminuit



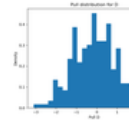
C_Fits_250.png
PNG 30KB



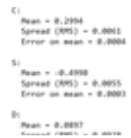
C_pulls.png
PNG 12KB



D_Fits_250.png
PNG 27KB



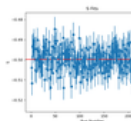
D_pulls.png
PNG 12KB



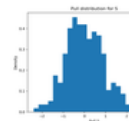
fit_CSD_values_250.png
PNG 11KB



pull_CSD_values_250.png
PNG 14KB



S_Fits_250.png
PNG 26KB



S_pulls.png
PNG 11KB

the fit plots are a bit crowded I'll sort that and generate some new ones, but other than that the pulls are not quite all consistent with 1 and 0, similar to Houde's and I'm gonna start on incorporating mistag rates soon

Houde Li Update your status [15:52](#)

I haven't done too much for the project in the last few days. In my previous fits, I only fitted $B \rightarrow f$. For some reason, my fit for $\bar{B} \rightarrow f$ is very bad despite using essentially the same code. Will work more on fitting both channels and fitting mis-tag rate in the next few days.

February 24

Houde Li  Update your status [10:50](#)

Fixed previous problem with $\bar{B} \rightarrow f$. Problem was I tried to scale the PDF normalisations directly instead of weighing the PDFs as coefficients.

Incorporated mis-tag rate ω as $PDF' = \frac{1}{(1-\omega)BF + \omega\overline{BF}} [(1-\omega)BF \cdot PDF + \omega\overline{BF} \cdot \overline{PDF}]$, where PDF and $\overline{PDF}$ are the correct/incorrect PDF for the particle, BF and $\overline{BF}$ are the branching fractions.

Result below is for the $B^0 \rightarrow J/\phi K_S^0$ channel. D was fixed to ≈ -0.7 and not fitted.

Pulls far more consistent after fitting both B and $\bar{B}$.

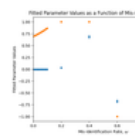
Fitted parameter values increases somewhat exponentially for small ω until reaching the parameter limit I set at 1. For large mistag $> 50\%$, the plot is roughly inverted (like a tan function).

Edit: fixed the expression for PDF', added the BF weightings to the numerator.

```
True parameter values
C_F = 0.000
S_F = 0.7500000000000000
C_F = 0.000

(20260527) Fitting results on
C_F: mean = 0.000000 ± 0.00
S_F: mean = 0.750000 ± 0.00
C_F: mean = 0.000000 ± 0.00
C_F_pull: mean = -0.000000 ± 0.000000
S_F_pull: mean = -0.000000 ± 0.000000
```

output.png
PNG 92KB



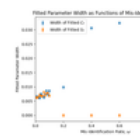
mistag.png
PNG 29KB

Matthew William Kenzie [12:12](#)

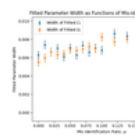
don't want the fitted parameter values as a function of omega but the uncertainty as a function of omega (width of bootstrapped distribution)

Houde Li  Update your status [14:57](#)

An overview and a zoomed in view of the width (and error in width) in C and S distributions for different ω . For S, the parameter value would max out for $\omega > 15\%$.



mistag_width_full....
PNG 31KB

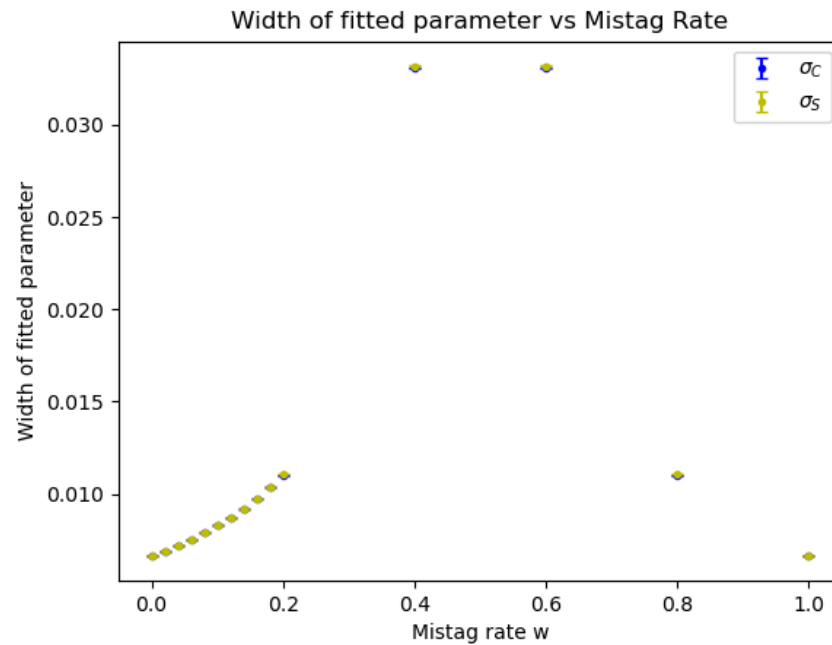


mistag_width_short.png
PNG 34KB

February 25

Asa Evan Kelsey 22:49

this is what mine has come out as with fixed $D \approx 0.999$, for $B_s \rightarrow J/\psi\phi$



I think the divergent behaviour at $w = 1/2$ is to be expected from what I've read?

February 26

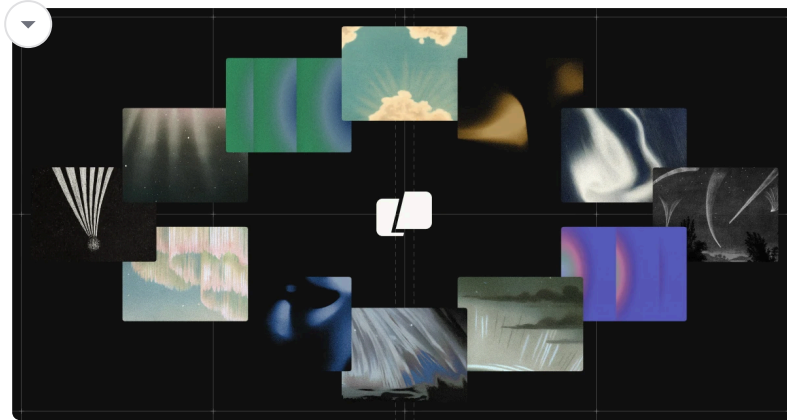
Matthew William Kenzie 10:16

<https://arxiv.org/pdf/2508.04471>

<https://www.warp.dev>

Warp: The Agentic Development Environment

Fast terminal, state-of-the-art agents, and cloud orchestration for the full software development lifecycle.

**March 06**

Asa Evan Kelsey 17:27

with the number of events i expect (around 10^7), trying to produce this curve again would take a completely unfeasible amount of time, so I'm not quite sure what to do about that

Asa Evan Kelsey 17:28

the initial one assumed around 50000 events, so this is a huge step up computationally

Asa Evan Kelsey 17:42

@Houde Li have you also tried to do it with the expected number of signals?

Houde Li  Update your status 17:45

Assuming an initial 6×10^{12} Z, I expect about 300M initial B^0 mesons for $B^0 \rightarrow J/\psi K^0$. I'm still working on other parts of my code at the moment, but I'm not optimistic about running this many.

Taking into account a minimum of 20% tagging power, the number of tag-able events is still quite big at 60M. For reference, I was doing 100K events in previous runs.

March 07

Asa Evan Kelsey [11:38](#)

yeah left it to run on my vm overnight and it is taking about 45 mins per toy run and i was previously doing 100 toys per value of ω , so this clearly isn't gonna work

Asa Evan Kelsey [11:46](#)

I'm gonna spend today trying to get it to a reasonable runtime with parallelisation and some other techniques and will get back to you about it

Houde Li  Update your status [13:29](#)

[@Asa Evan Kelsey](#)

I have a few concerns about this. It takes 45 min to do one run, 100 runs per ω , around 10 ω per plot, then we need (I remember?) a set of several such plots. Even if we had the maximum of 64 cores offered by the department, I still don't think we can reduce the time to a reasonable level.

What I think might be a huge improvement is to bin the PDF, do a Poisson on each bin about the expected number of events in it given the several million events, then feed the bins to the binned cost function. So instead of generating millions of events each run, we can basically just generate say 100 bins over 5 Γt .

I'm not sure what implication this will have on the accuracy of the result, but it will definitely be faster.

Asa Evan Kelsey [13:46](#)

Yeah that was one thing I was planning on doing too

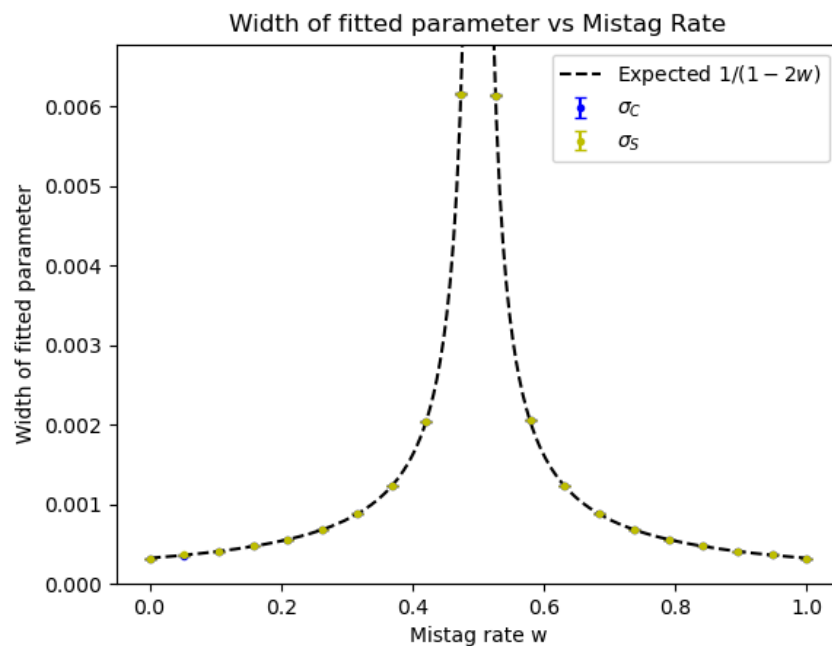
March 08

Asa Evan Kelsey [11:35](#)

To be honest, I feel like the obvious fix here is still using maybe around 50-100k, but then just scaling our results to N_s ? i don't really see any reason why this wouldn't work.

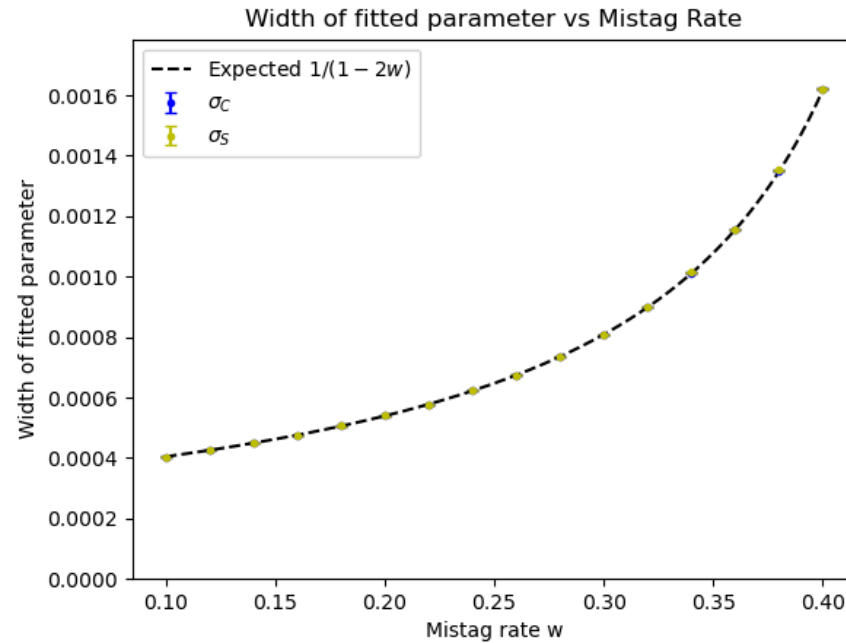
Asa Evan Kelsey [12:09](#)

I've managed to speed my code up massively using exponential sampling instead of rejection sampling, and have gotten this plot with 500 toys and about 30000 events which only took about 5 mins to produce, then scaled it to the expected number of events



is there anything to be said about using too many toys as it can reduce the make the error arbitrarily small (albeit slowly) by just doing more and more toys?

this is it on a more reasonable range



Houde Li  Update your status 16:33

I'm still working on incorporating mis-tag. I think the code is correct, but the width vs mis-tag plot is very linear. Not sure if that's a problem.

The limiting factor seems to be iminuit's fitting. In comparison, the time to generate events is negligible.

Asa Evan Kelsey 16:44

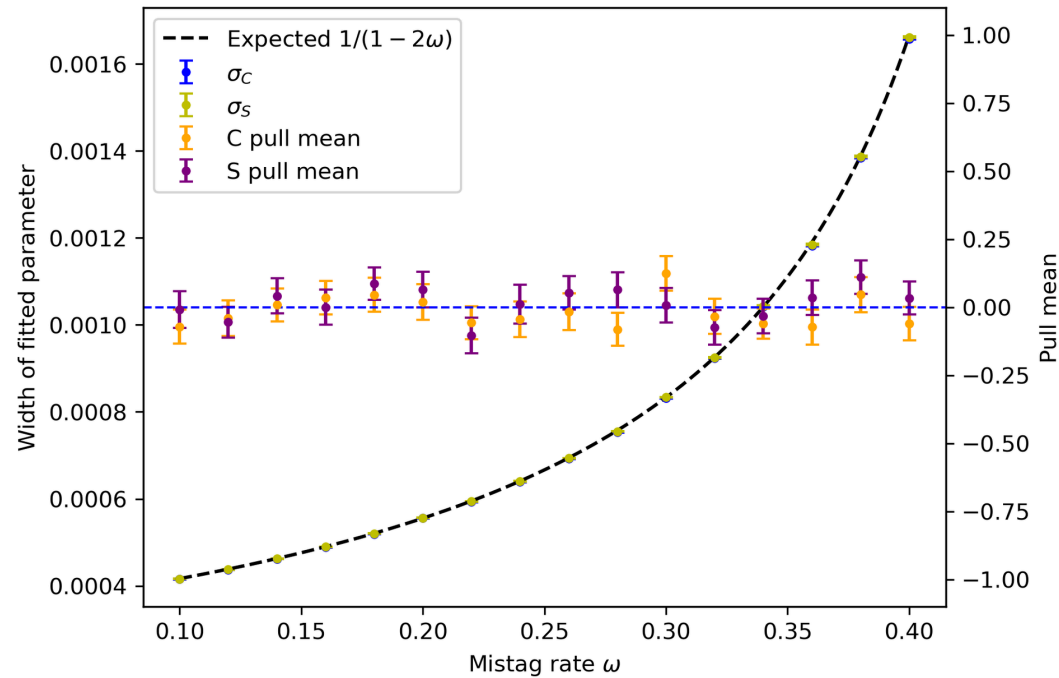
I'm pretty sure the plot should be $\frac{\sigma_0}{1-2\omega}$ from page 1 of this paper <https://arxiv.org/pdf/hep-ex/0009058>



can send over bits of my code if needed

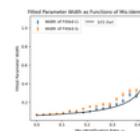
Asa Evan Kelsey [19:35](#)

got this plot too which has the pulls, which are mostly consistent with 0 but there may be a little bit of bias in places

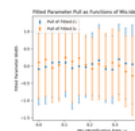


Houde Li [Update your status](#) [21:52](#)

I got my width vs mistag ready as well. The sample size is 1000.



width.png
PNG 35KB



pull.png
PNG 28KB

March 09

Matthew William Kenzie [13:09](#)

This is lovely stuff. Can you put y axis units into degree error on beta or radian error for phis

Then add horizontal line at world average current values

Houde Li  Update your status 14:09

Do we need to account for the change in tagging power (and hence the taggable sampled size) as we vary the mis-tag rate?

Matthew William Kenzie 14:12

yes

but that should kind of be backed in

you still have the same sample size

it's just how you tag them

i think you could rewrite mistag rate in terms of effective sample size

Houde Li  Update your status 14:14

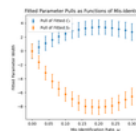
And then we rescale the width and error in width according to the taggable sample size?

Houde Li  Update your status 14:56

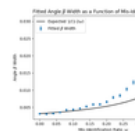
Fits for β using 100k initial mesons and 250 runs.

Width of β scaled to 600M initial mesons.

Current world average uncertainty is 0.45° , which is too big to put on the scaled width plot.



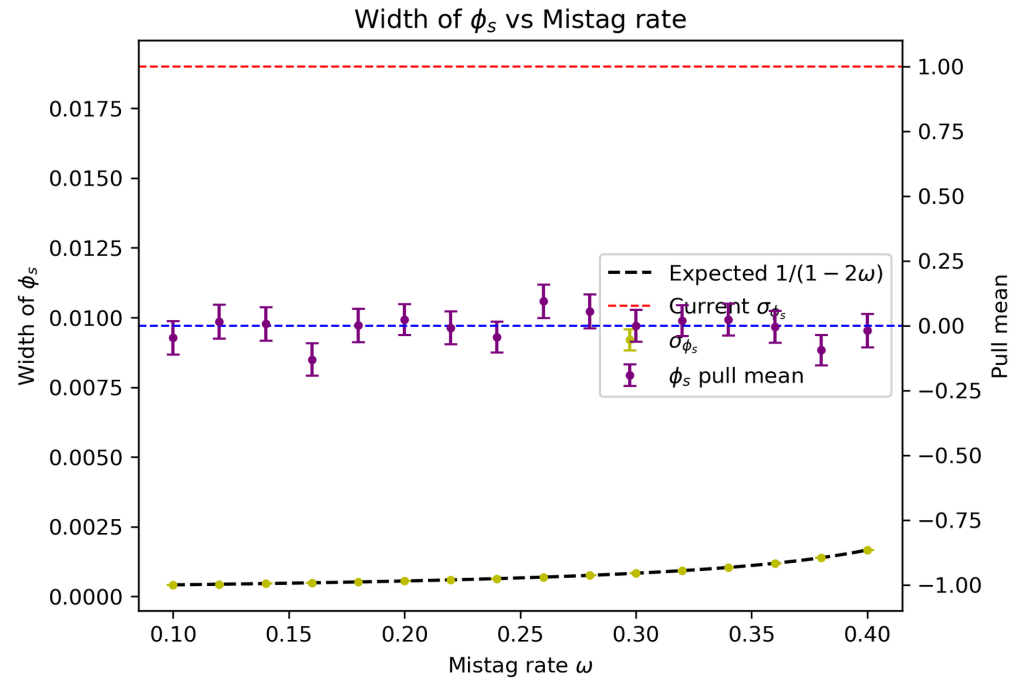
parameter_pull.png
PNG 28KB



beta_width.png
PNG 31KB

Asa Evan Kelsey [15:10](#)

obviously I'll move the legend, but the projected width with the number of events at FCC is far lower than the current value



red dashed line is the current precision at 0.019, bottom points are expected FCC precisions at different ω

I guess it makes sense, they wouldn't be building it otherwise

left y-axis is in radians for reference, I'll add that too

Houde Li  Update your status 09:39

There are still a few things I'm not quite sure about:

How does branching fraction work when B^0 and $\bar{B}^0$ can mix into each other and have different decay rates to f ? The "instantaneous" $t=0$ decay rates would be very close as C is close to zero, but the time-integrated branching fraction (integrate decay rate from $t=0$ to infinity) has a ratio of 2:1 for $\bar{B}^0:B^0$.

Despite B^0 and $\bar{B}^0$ being produced equally from Z , do we need to weigh the expected number of $B^0/\bar{B}^0 \rightarrow f$ events by the time-integrated branching fraction when sampling from Poisson?

Should the particles tagged as B^0 be fitted to $(1 - \omega)\text{PDF}(B^0) + \omega\text{PDF}(\bar{B}^0)$ or

$\frac{(1-\omega)\text{DecayRate}(B^0) + \omega\text{DecayRate}(\bar{B}^0)}{(1-\omega)\text{BF}(B^0) + \omega\text{BF}(\bar{B}^0)}$, where BF is the time-integrated branching fraction, $\text{DecayRate} = \text{BF} \times \text{PDF}$, and $\int dt \text{PDF} = 1$.

March 22

Houde Li  Update your status 12:16

Update:

My pull distribution after incorporating mis-tag is not consistent with $N(0,1)$. Will try to fix.

Still working on the background fitting, should get some result soon.

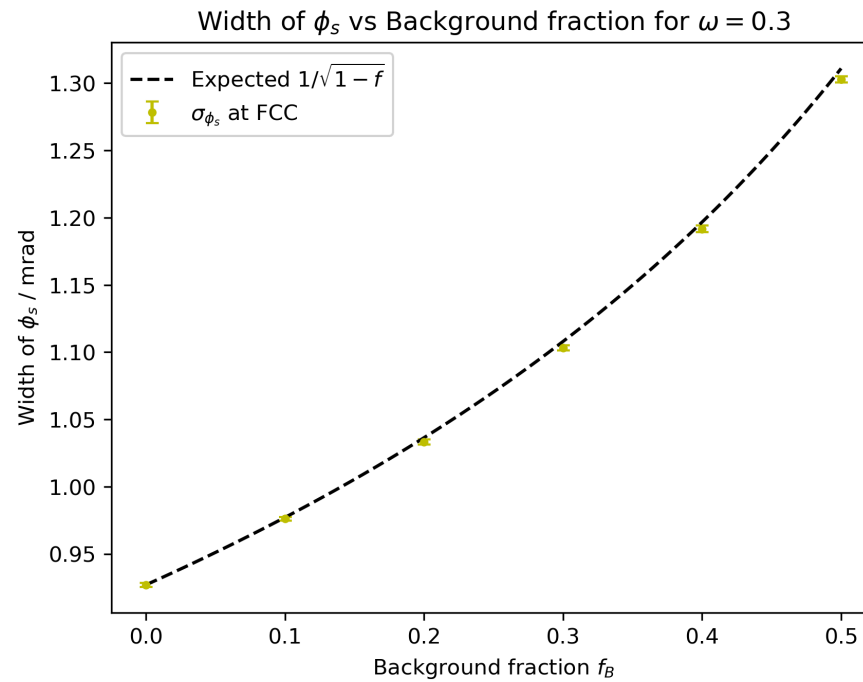
Statistics for the parameter values and pulls might need extra work. I think first order approximations aren't accurate for large width and errors in width and mean. This would affect the results near 50% mis-tag.

I tried to do "ls /eos/user/m/mkenzie/fcc/kenzie_CPV_tuples_no_selection" on SWAN to have a look at the simulated data, it said operation not permitted. Is this a permission issue?

March 23

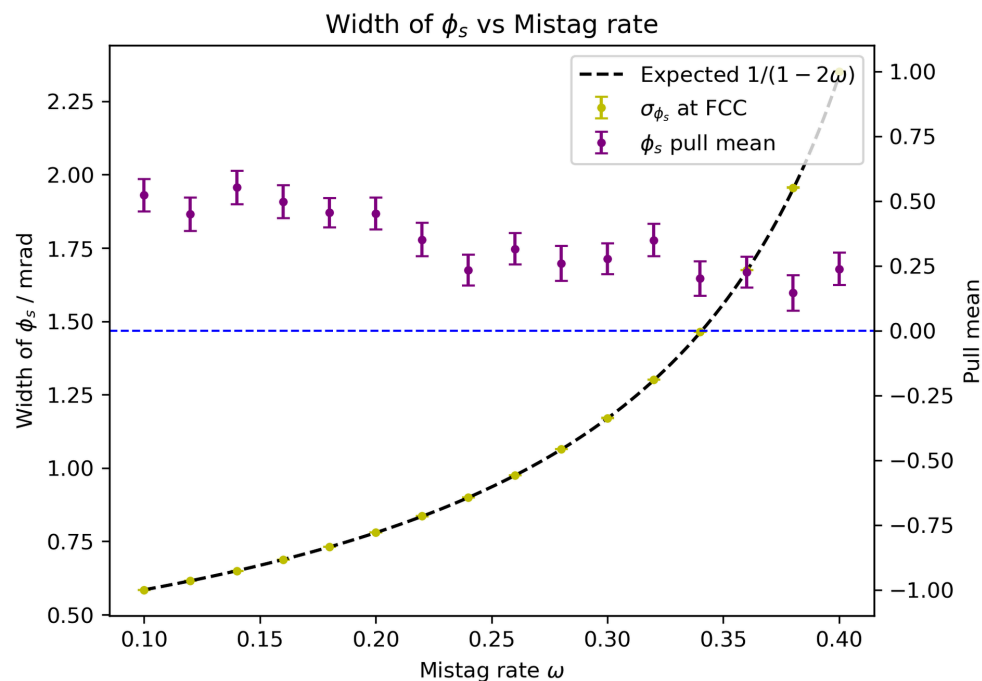
Asa Evan Kelsey [16:45](#)

I've managed to get something like this, although I imagine that a more relevant plot would be varying mistag rate for a fixed background fraction obtained from the simulations?



Asa Evan Kelsey [20:13](#)

managed to get this at $f=0.2$, which has larger y-values which is good and makes sense, but seems a bit biased



was also struggling with bias for the plot vs f

Matthew William Kenzie [21:09](#)

this one does indeed look biased which suggests an issue. Just to check when you inject background you also fit back for background?

Asa Evan Kelsey [21:36](#)

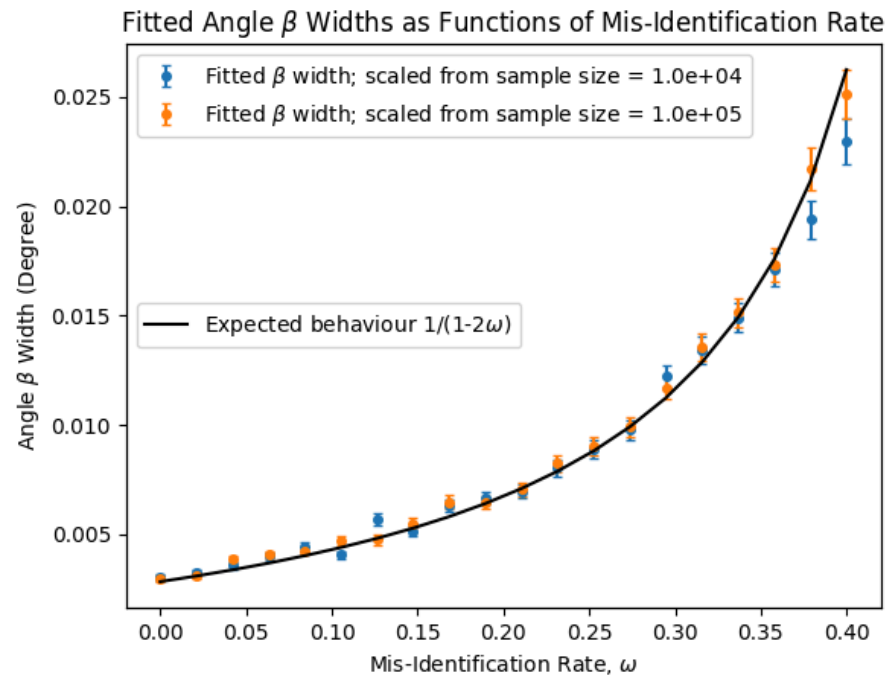
haven't been including f as a fitted parameter in either case no, just fixed it for second plot and varied it between 0 and 0.5 for first

Asa Evan Kelsey [22:14](#)

have managed to fix the bias on the second plot

Houde Li  Update your status [22:29](#)

Plot of angle β width vs mis-tag rate. No background here. Scaling to the expected number of signals (600M) works consistently with a few sample sizes I tried (10K and 100K).



I still have some problems with pull when trying to incorporate background, should be able to resolve them soon.

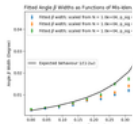
Matthew William Kenzie [22:33](#)

lovely plot this

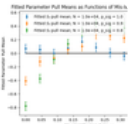
March 24

Houde Li  Update your status 03:28

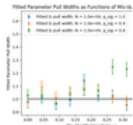
Plot of angle width vs mis-tag rate. This looks okay, but the pull distribution is still bad.



angle_width_vs_...
PNG 46KB



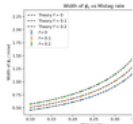
pull_mean_vs_mistag_...
PNG 40KB



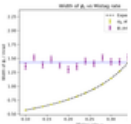
pull_width_vs_mistag_...
PNG 40KB

Asa Evan Kelsey 12:23

just gotten this plot, pulls are mostly consistent with 0 as shown in single plot



mistag_errors_FC...
PNG 159KB



mistag_errors_FCC_f=0...
PNG 144KB

Matthew William Kenzie 14:40

Back of the envelope: B0 -> Jpsi KS Ns = 14M, Nb = 36M

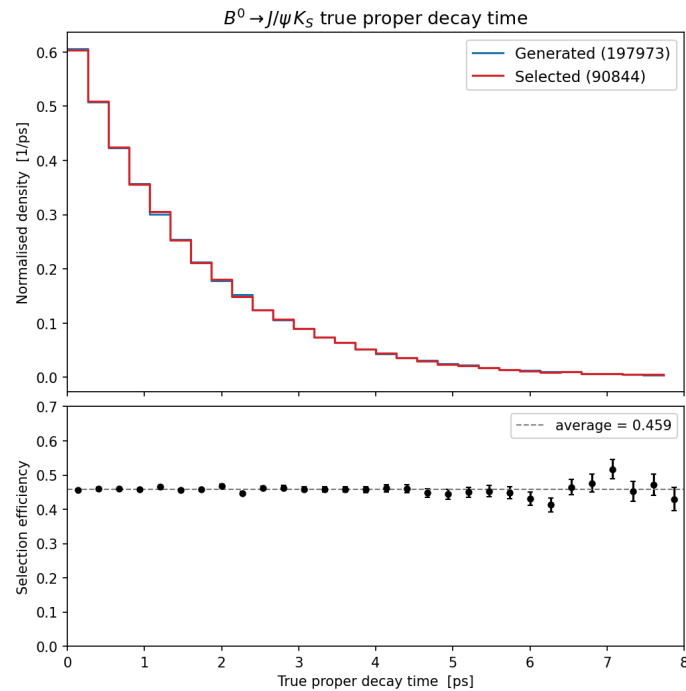
Matthew William Kenzie [14:48](#)

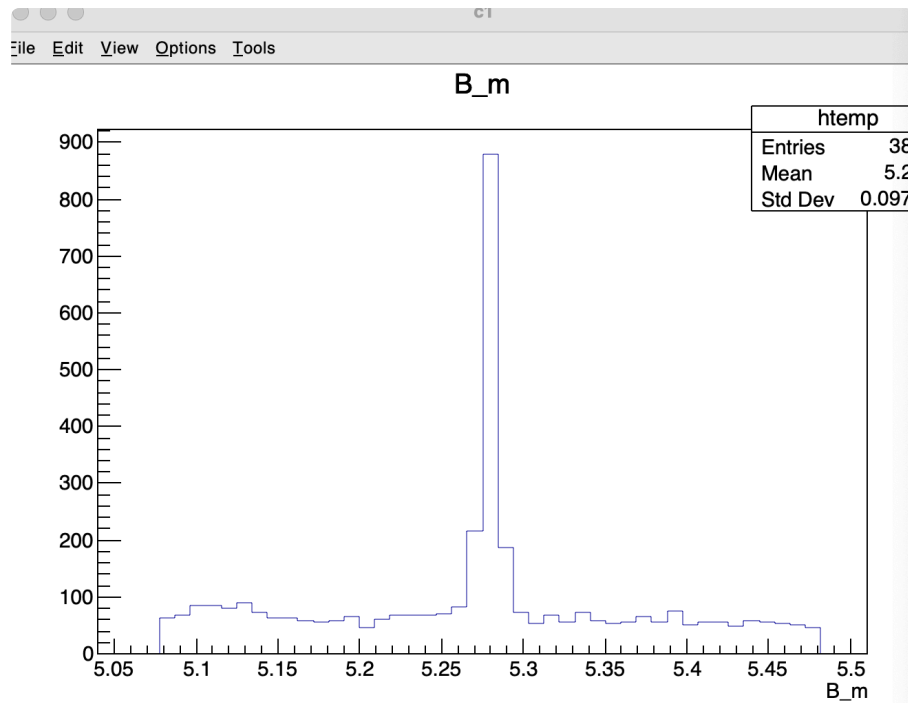
signal fit of selected candidate lifetimes:

```
FCN=45.5954 FROM MIGRAD   STATUS=CONVERGED   41 CALLS   42 TOTAL
                        EDM=1.68003e-09   STRATEGY= 1   ERROR MATRIX ACCURATE
EXT  PARAMETER
NO.   NAME      VALUE      ERROR      STEP      FIRST
      NAME      VALUE      ERROR      SIZE      DERIVATIVE
  1 Constant    9.17964e+00  4.77459e-03  1.09636e-05  1.70830e-02
  2 Slope       -6.51959e-01  2.32012e-03  5.32751e-06  3.16133e-02
```

background fit of selected candiare lifetimes:

```
FCN=65.9356 FROM MIGRAD   STATUS=CONVERGED   46 CALLS   47 TOTAL
                        EDM=1.18712e-07   STRATEGY= 1   ERROR MATRIX ACCURATE
EXT  PARAMETER
NO.   NAME      VALUE      ERROR      STEP      FIRST
      NAME      VALUE      ERROR      SIZE      DERIVATIVE
  1 Constant    5.07891e+00  3.29775e-02  8.52805e-05  2.25637e-02
  2 Slope       -4.44530e-01  1.24653e-02  3.21779e-05  3.96615e-02
```





Matthew William Kenzie [15:01](#)

For $B_s \rightarrow J/\psi \Phi$: N_s (with 100% efficiency) is 10M, $N_b = 20M$
 mass windows are ± 100 MeV

May 03

Asa Evan Kelsey [16:03](#)

for adding secondary vertex branching factors, does that mean that the number of signals is not actually 10M, or does the 10M already include this?

like do i need to multiply 10M by $(BR(J/\psi \text{ to } e^+e^-) + BR(J/\psi \text{ to } \mu^+\mu^-)) \times BR(\phi \text{ to } K^+K^-)$?

Houde Li [18:23](#)

I think 10M would be the number of events you are left with after the full selection, which should already include conditioning on final state decay modes.

May 04

Matthew William Kenzie [07:47](#)

indeed as **Houde** says that already includes the intermediate branching fractions

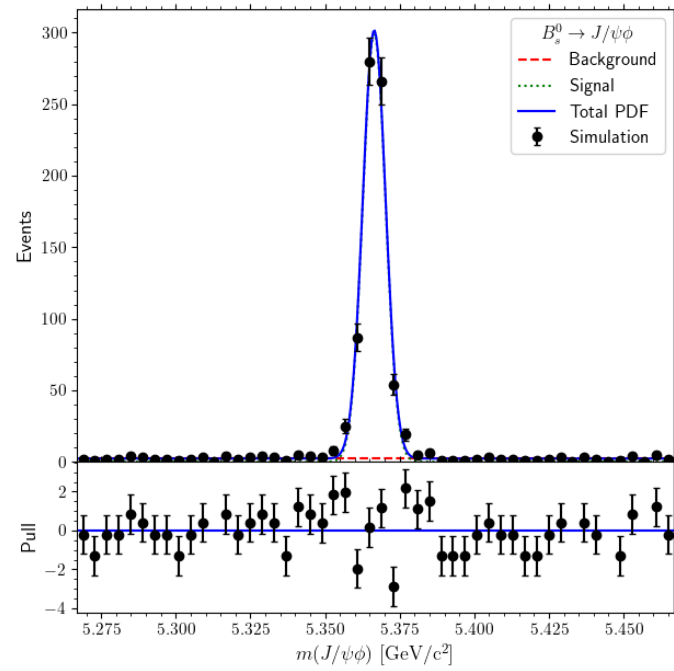
[@Asa Evan Kelsey](#) . I wondered if you could actually run a slight alternative for me. I ran some specific studies on the Bs->JpsiPhi mode and got to expected numbers a bit different. suggesting 8M signal vs 2M background. the result of my studies i put below for both modes (which show rates in a wide window and then compute them for a slightly narrower window)

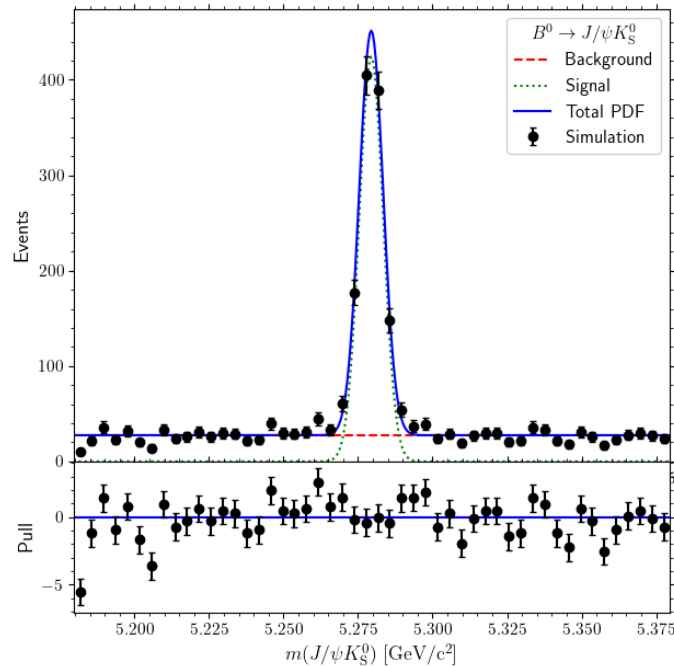
Bs Mode

```
Mass window:      [5.26691,5.4669099999999995]
Nbkg in window:   114.74120455059729
Nsig in window:   716.4567964978469
Bkg eff in window: 2.2036811385687696e-06
Sig eff in window: 0.7455638
With this selection and 6e12 Z bosons
--> Expected N signal:      8,386,102
--> Expected N background:  1,999,180
```

Bd Mode

```
Mass window:      [5.17966,5.3796599999999999]
Nbkg in window:   1375.1308052860745
Nsig in window:   1065.9777013805701
Bkg eff in window: 2.641030053490416e-05
Sig eff in window: 0.46988225
With this selection and 6e12 Z bosons
--> Expected N signal:      1,316,737
--> Expected N background: 23,959,425
```

BS fit plot

Bd fit plot

Matthew William Kenzie 08:03

for a next meeting i could do Thursday afternoon if that would be helpful

one final point on **flavour tagging** metrics. At the moment we have been using the mistag rate, ω , as the main metric. there is another quantity in tagging called the tagging **efficiency** (we mostly assume that this will be 100). then there is a quantity called the "dilution" which essentially is a multiplicative factor which "dilutes" the TD CP asymmetry sensitivity which goes like:

$$D = (1 - 2\omega)$$

and then the tagging power is

$$P = \epsilon D^2 = \epsilon(1 - 2\omega)^2$$

the tagging power we expect to be somewhere around 30-40% (which with an efficiency of close to 100%) would give a mistag rate of somewhere around 20%

Asa Evan Kelsey [11:11](#)

with 8M sig 2M bkg i've got a width on ϕ_s of about 1.26 mrad which is short of the literature hoped for value of 4.6 mrad but I suppose that's why we are doing this

Asa Evan Kelsey [11:11](#)

yeah that works for me

Asa Evan Kelsey [11:16](#)

also the 8M and 2M have the efficiencies included in those right?

Houde Li  Update your status [12:43](#)

works for me as well

Houde Li  Update your status [12:48](#)

I have a question about background events. How is Nbkg determined?

I would think Nbkg would contain all events in non-signal channels that might be reconstructed as signal events due to errors, but this would need an acceptance of bkg events. Is the quoted Nbkg the number of "accepted" background events?

Matthew William Kenzie [12:54](#)

Exactly that. Number of background that makes it through acceptance and selection criteria

Will need to check what window they are in

Asa Evan Kelsey [19:08](#)

Just one small thing about my recent work - you asked me to float the bkg fraction in my minuit fitting, but that has introduced a statistical bias that I have spent the better part of a day trying to solve before calling it quits as I don't feel the minor change is really worth it - is that something that I should absolutely positively include or was I justified in just leaving f_{bkg} fixed in the minuit fit

Matthew William Kenzie [20:11](#)

ok to leave it fixed if its causing an issue

but i think if it's fixed then there's not much point in including background

Asa Evan Kelsey [20:52](#)

yeah i suppose but when i increase the fixed background fraction it does increase the width as expected due to it affecting the pdf sampling with behaviour as expected it just isn't biased anymore i'm finding it quite strange

Houde Li  Update your status 21:08

[@Asa Evan Kelsey](#) I suspect you would be fixing the total number of events, right? It might be worth a try to fix the number of signal events instead.

Asa Evan Kelsey 21:11

ah ok thanks will give it a try

Houde Li  Update your status 21:11

I think fitting the bkg fraction touches on fitting the number of events, which requires using extended maximum likelihood fits. I don't know about you, but I wrote my cost function in such a way that its difficult to implement this. But if you just fix N_{sig} and float bkg fraction, any bias due to fitting the event number should go to the bkg fraction and should not affect S.

Houde Li  Update your status 12:43

works for me as well

Houde Li  Update your status 12:48

I have a question about background events. How is Nbkg determined?

I would think Nbkg would contain all events in non-signal channels that might be reconstructed as signal events due to errors, but this would need an acceptance of bkg events. Is the quoted Nbkg the number of "accepted" background events?

Matthew William Kenzie 12:54

Exactly that. Number of background that makes it through acceptance and selection criteria

Will need to check what window they are in

Asa Evan Kelsey 19:08

Just one small thing about my recent work - you asked me to float the bkg fraction in my minuit fitting, but that has introduced a statistical bias that I have spent the better part of a day trying to solve before calling it quits as I don't feel the minor change is really worth it - is that something that I should absolutely positively include or was I justified in just leaving f_{bkg} fixed in the minuit fit

Matthew William Kenzie 20:11

ok to leave it fixed if its causing an issue

but i think if it's fixed then there's not much point in including background

Asa Evan Kelsey 20:52

yeah i suppose but when i increase the fixed background fraction it does increase the width as expected due to it affecting the pdf sampling with behaviour as expected it just isn't biased anymore i'm finding it quite strange

Houde Li  Update your status 21:08

[@Asa Evan Kelsey](#) I suspect you would be fixing the total number of events, right? It might be worth a try to fix the number of signal events instead.

Asa Evan Kelsey 21:11

ah ok thanks will give it a try

Houde Li  Update your status 21:11

I think fitting the bkg fraction touches on fitting the number of events, which requires using extended maximum likelihood fits. I don't know about you, but I wrote my cost function in such a way that its difficult to implement this. But if you just fix N_{sig} and float bkg fraction, any bias due to fitting the event number should go to the bkg fraction and should not affect S .

May 05

Asa Evan Kelsey 11:13

Are we expected to be at the presentations all day tomorrow or just for our block? I'm not sure I can really afford to take 7 hours out of my day for all the presentations.

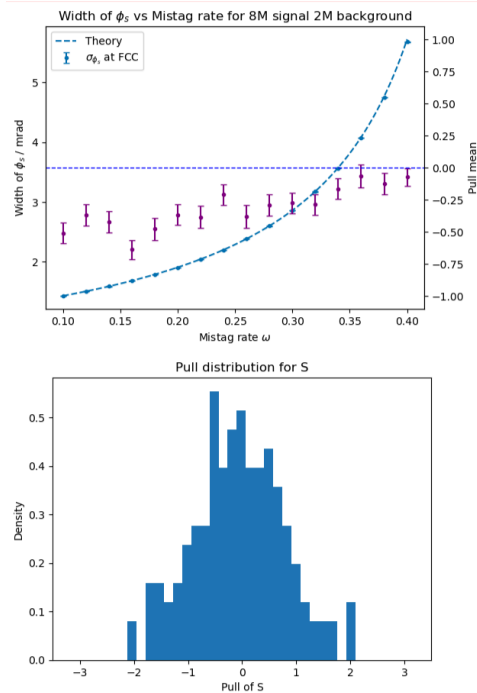
Matthew William Kenzie 12:36

just your own block will be fine.

obviously helps others not to be presenting to an empty room but i think within your block will be totally fine

Asa Evan Kelsey 16:42

These are my results - the pulls look biased in the first figure, but the histogram doesn't actually look that biased to me? This is using your suggestion **Houde**



this is with floated background

Houde Li  Update your status 16:43

Is that the pull of the angle ϕ ? @Asa Evan Kelsey

Matthew William Kenzie 16:43

Histogram indeed looks fine. Maybe there are outliers skewing the mean

Asa Evan Kelsey 16:44

ϕ_s yeah i forgot to change the title

Asa Evan Kelsey 16:44

will have a look over a bigger range

Houde Li  Update your status 16:45

If S is gaussian distributed, I don't think we would expect arcsin(S) to be gaussian distributed, hence pull of phi not being exactly consistent with N(0, 1) should be expected to some degree.

Perhaps plot pull of S?

Asa Evan Kelsey 16:48

will run that now

Houde Li  Update your status 16:50

Wait but your angle is zero, so that shouldn't distort things too much ..

Asa Evan Kelsey 16:52

its pretty close to it yeah - i suspect the problem will be big outliers once i increase the xlims

Houde Li  Update your status 16:52

Did you look at pull distribution plot of S at other omega? There's no way you have a non-zero pull mean but it does not show on the plot.

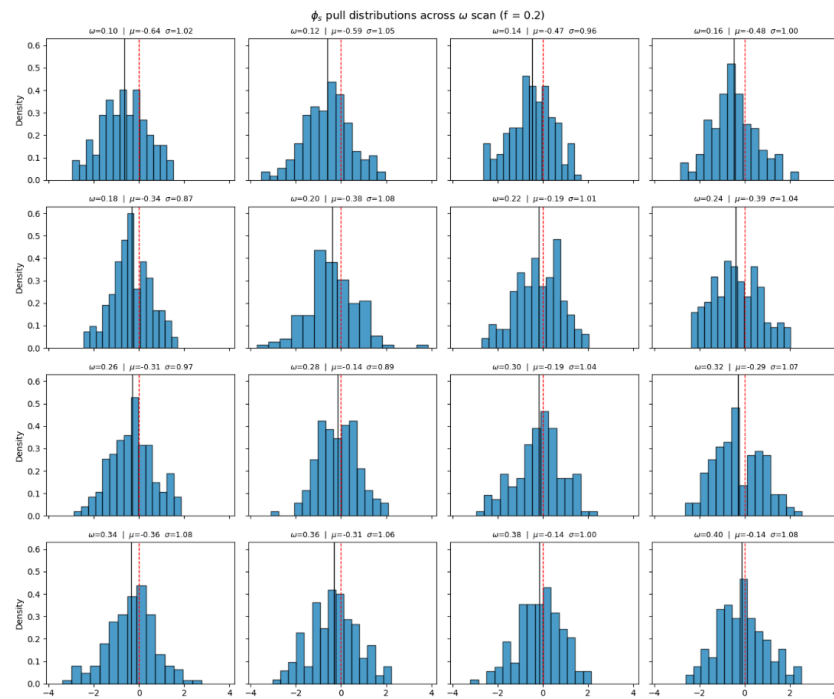
Asa Evan Kelsey 17:00

having a look at it now

i think that histogram only looks so normal due to it being the highest omega val

Asa Evan Kelsey 17:13

yeah these are definitely just skewed to the left...



May 07

Houde Li Update your status 10:16

I was looking at the derivation of $S = \sin 2\beta$. However, using my result, which I'm still not sure if/how it's wrong, I would get $S = -\sin 2\beta$...

Matthew William Kenzie 12:17

i think you are correct and this is because the $J/\psi K_S$ final state is CP-odd so has $\eta = -1$. For $J/\psi K_L$ (which is CP-even) one has $\eta = 1$

there is some discussion around P43 of these notes: <https://www.nikhef.nl/~h71/Lectures/2020/ppII-cpviolation-14022020.pdf>

(but the literature also sometimes has different sign conventions)

the general formula i think would provide:

$$S_f = -\eta_f \sin(2\beta)$$

and thus for $J/\psi K_S^0$ where $\eta_f = -1$ you get $S_f = \sin(2\beta)$

Houde Li  Update your status 12:24

I re-did the derivation this morning, this time including the CKM elements. I believe I used the exact same convention as the PDG paper. Regardless, I still seem to be getting the incorrect physical result.



diagonalisation_2....
PDF 1MB

I accounted for $\eta_f = -1$

The problem is still the expression of q/p in terms of the CKM elements.

Perhaps I'm mistaking the interpretation of H_{12} ? That should be the interaction of $B^0 \rightarrow \bar{B}^0$, I think?

Matthew William Kenzie 12:30

shouldn't you have $e^{-i2\beta}$

rather than $e^{i2\beta}$?

Houde Li  Update your status 12:32

It's $\beta = -\arg V_{tb}^* V_{td}$, phase of lambda is $V_{td}^* V_{tb}$.

the b and d subscripts are switched

Matthew William Kenzie 12:39

i think your q/p needs a minus sign in the exponent

but not quite sure it enters

Houde Li  Update your status 12:40

This paper states $M_{12} \propto (V_{tb}^* V_{td})^2$ corresponding to the $B^0 \rightarrow \bar{B}^0$, which is different from my interpretation.

Matthew William Kenzie 12:40

depends then whether you choose to start with a B^0 or $\bar{B}^0$ i think

^{1 2 3 4}
sometimes you see $q/p \approx \sqrt{M_{12}^*/M_{12}}$ - what happens if you use that instead?

Houde Li  Update your status [12:42](#)

But I would think $id/dt |B^0\rangle \ni H_{12} |\bar{B}^0\rangle$ means its a contribution of $\bar{B}^0$ to B^0 , so it's the $\bar{B}^0 \rightarrow B^0$ diagram?

Houde Li  Update your status [12:44](#)

This would give the correct result if I use $M_{12} \propto (V_{td}^* V_{tb})^2$

Matthew William Kenzie [12:48](#)

mmm... ok. are you 100% sure you have (q/p) the right way around (bottom of page 1), you conclude that

$$\left(\frac{q}{p}\right)^2 = \frac{H_{12}}{H_{21}}$$

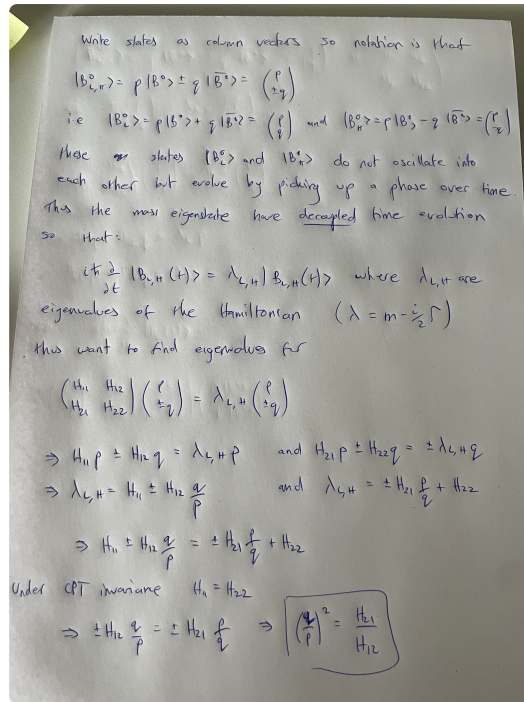
I wonder if that should be H_{21}/H_{12} . can you check that?

Houde Li  Update your status [12:52](#)

This is in accordance with the PDG paper. I haven't swapped p/q. I assumed H_{11}/H_{22} are for $B^0/\bar{B}^0$, which I can't find if it's stated in the paper right now.

Matthew William Kenzie 13:18

I think it's somewhere where you define S which is actually the transpose of the matrix you want



You transpose H when you shouldn't I think

Houde Li 🙄 Update your status 13:24

This is essentially what I wrote in the middle of page 1 of my work, where I defined the mass eigenstates as column vectors of p and q, but without a transpose I believe you would be labelling B0 and B0bar as 2 and 1 respectively

Houde Li 🙄 Update your status 13:52

Using the normal eigenvalue solution this way does not recover the correct subscripts



scan.pdf
PDF 479KB

The PDG paper did not state if subscript $1/2$ represent $B_0/B_0\text{bar}$, but two of its references on this topic did write out the whole matrix.



1-s2.0-0550321381...
PDF 983KB



PhysRevD.23.1567.pdf
PDF 722KB

Matthew William Kenzie [15:47](#)

this is where the projection for beta comes from:

<https://inspirehep.net/files/56aa1f400a6312a57c03e5f693c7667e>

Take a look in Table 7.3 it says $\sin(2\beta)$ will be measured with an uncertainty of 0.005 which equates to approx 0.18 degrees

Houde Li Update your status [16:05](#)



Meeting_Notes.pdf
PDF 2MB

May 08

Asa Evan Kelsey [14:02](#)

@Houde Li i am writing up my mistag portion currently, what numbers are you using for tagging power and efficiency? My 22% came from a 95% efficiency and 30% tagging power, which I thought seems reasonable but I'm struggling to find literature for this

Houde Li Update your status [14:04](#)

Unfortunatly I haven't had the time yet to look at what mistag rates we are expecting in the literature. I'll try to have a look over the weekend.

Matthew William Kenzie [15:11](#)

very little in the literature that will get you these directly

you could use LEP flavour tagging as the baseline (which is around 20% power). Belle-II have recently got something called the TFLaT algorithm which can do around 40% tagging power (with about 95% efficiency)

you could hunt for a reference to that

Asa Evan Kelsey [11:49](#)

are we ok to use these in our reports if we cite a pers comm?

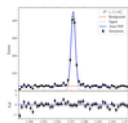
Matthew William Kenzie [15:19](#)

yes absolutely

i was going to make some fixes to them

so let me do that

This is for the Jpsi KS



mass_fit.png
PNG 41KB

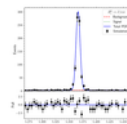


mass_fit.pdf
PDF 257KB

This is for the Jpsi Phi



mass_fit.pdf
PDF 289KB



mass_fit.png
PNG 38KB

May 15

Houde Li [Update your status](#) [12:53](#)

Should we plot the error in S against dilution or dilution squared (which would give a straight line)?

Matthew William Kenzie [15:30](#)

doesn't matter

Houde Li [Update your status](#) [23:12](#)

Found the problem: in my definition of the mass eigenstates, the basis states were time dependent.

May 20

Houde Li [Update your status](#) [11:00](#)

For the Bd mode, I believe there's a typo right? Where the expected number of signal events is 13M instead of 1.3M?

Houde Li  Update your status [11:05](#)

Also, was the selection done based on truth-level final states (e.g., $\pi^+\pi^-$ for the reconstruction of K_S^0), or were the final states reconstructed from simulated detector response?

May 21

Matthew William Kenzie [11:23](#)

i don't think it's a typo

it's what my code spits out so if you think there's a mistake we should try figure out where

Matthew William Kenzie [11:29](#)

hang on so maybe there is a bug in my code

it essentially prints this

```
B mass mean:          5.279408517984122
B mass width:         0.004011223060802155
Mass window:          [5.07966,5.47966]
Nbkg in window:       2750.2616105721613
Nsig in window:       1065.9777013805701
Bkg eff in window:    5.2820601069808556e-05
Sig eff in window:    0.46988225
With this selection and 6e12 Z bosons
--> Expected N signal:      1,316,737
--> Expected N background: 47,918,849
```

Asa Evan Kelsey [11:32](#)

Are the B_s numbers fine?

Matthew William Kenzie [11:32](#)

so in some region I find $N_b = 2750$ and $N_s = 1065$ (this is for a $\pm 200\text{MeV}$ window) this suggests a S/B ratio of $\sim 3:1$.

However when i try to use this to calculate the efficiency and then apply this to known BF's etc I actually find expected N_{sig} 1.3M (roughly 1000 larger) and 48M background (which does not scale). now my S/B ratio is like 48:1.

i think i'm somehow scaling the background number wrongly.

Matthew William Kenzie [11:38](#)

the error is here somewhere in my script:

```
### calculation ###
eff = tree.arrays("efficiency")["efficiency"][0]
nevs = len(mass)
ns = mi.values["Ns"]
nb = mi.values["Nb"]

bkg_eff = eff * nb / nevs
```

the `eff` number is for a mixture of signal and background so i want to extrapolate that to the background only efficiency

Matthew William Kenzie [11:50](#)

found the bug. it was in the branching ratio further up

i was an order of magnitude off

so yes should be 13M signal and 24M background

for the JpsiPhi it's unchanged

should be 8M signal and 2M background

for the Flavours @ FCC workshop i have found a solution where another speaker (PhD student at Oxford is going to present some stuff they have been working on with neutral modes). he will then present a few slides on our behalf with the work on these "golden modes" - this then avoids me speaking as the convener.

but still gets our work shown

Houde Li  Update your status [11:52](#)

Is this based on truth-level final state particles?

Matthew William Kenzie [11:53](#)

sort of yes sort of no

it's reconstruction information

but some of that is seeded from the truth (for speed) but it wouldn't be in real life data

here is the fit.py script by the way if i've never shared it



fit.py
PY 6KB

Houde Li Update your status [11:56](#)

So right now I do have the estimated error on β given a range of tagging power. Is a way to quote an error due to possible difference in selection efficiency?

Matthew William Kenzie [11:56](#)

not sure i understand the question

[11:57](#)

yes there is a way to quote due to difference in selection efficiency but you would also need to quantify what change in the S/B ratio you get as the efficiency changes

Houde Li Update your status [11:58](#)

Yes, what I mean is basically do we know what range of signal fraction we can expect instead of just having a single number?

Would it also be more precise to call what we have been calling the "acceptance" the "selection efficiency" instead?

Matthew William Kenzie [15:52](#)

it's both

Today

Asa Evan Kelsey [12:57](#)

Hi Matt, is there any chance that you could export the chat logs? We need to submit something for the lab notebook and I'm mindful of the fact that the submission is on a Monday and you won't be reachable over the weekend. Thanks in advance!

Asa Evan Kelsey [14:34](#)

Also, when will we know whether we are poster or oral presenters?

Matthew William Kenzie [14:37](#)

I don't have admin rights to do this.

Download chat history.

Perhaps you could provide a link and then download from web interface directly?